Hidden Tails: Certifying Tail-Risk Claims under Selective Labels

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Abstract

Selective labels can hide the losses that determine a predictor's population tail risk. Even unlimited historical data can leave its conditional value-at-risk (CVaR) unresolved. For binary outcomes and bounded losses, we derive the sharp population CVaR interval under an outcome-dependent odds-ratio sensitivity model: every value is compatible with the observations and the stated restriction. We construct calibrated statistical intervals enclosing every compatible risk simultaneously across candidate predictors at fixed sensitivity and tail levels, using cross-fitted bias correction. These certificates separate persistent hidden-outcome ambiguity from sampling error and support risk limits, predictor comparisons, and conservative selection. We also derive the minimum-cost revelation allocation for a fixed predictor and loss threshold. Tail identification value measures ambiguity removed when audited contributions become known and the original sensitivity bounds apply separately to audited and still-hidden outcomes. Synthetic and lending experiments examine endpoint accuracy, population assessment, and targeted audits. Under controlled masking of known repayment outcomes, revealing about 16% of hidden labels removes a median 61% of estimated fixed-threshold ambiguity, versus 16% for random revelation. The framework specifies which tail-risk claims the evidence supports and where additional labels improve precision.

1 Introduction

Population tail-risk claims guide decisions about whether a predictor meets an acceptable loss limit and whether one predictor has lower tail risk than another. Yet historical decisions determine which outcomes are available for evaluation [15–17], and the missing cases can carry the largest prediction losses. For example, a hospital assessing a model for 30-day adverse events may lack event status for patients who receive subsequent care outside its network; record linkage can later recover those labels. Conditional value-at-risk (CVaR) averages the upper tail of prediction loss [1, 2, 28]. Certifying this quantity for all target patients therefore requires an explicit account of hidden outcomes.

The obstacle persists even if the observed distribution is known exactly. Suppose labels are observed for 25% of a population with constant covariates, with equal proportions of zeros and ones among observed outcomes. Under zero-one loss and CVaR at $\alpha = .5$, the constant predictors $f = 0$ and $g = 1$ have identical observed loss distributions. If every hidden label is zero, their population CVaRs are $(.25, 1)$; if every hidden label is one, they are $(1, .25)$. Both worlds produce the same records and have positive observation probability. This illustration allows unrestricted hidden outcomes and shows why historical records alone cannot certify an ordering (Appendix A.2).

We make the assumptions needed for certification explicit through a finite, externally specified sensitivity level Γ, which limits how much outcome odds can differ between hidden and observed cases at the same covariates. The compatible population laws yield a sharp CVaR interval: every value in

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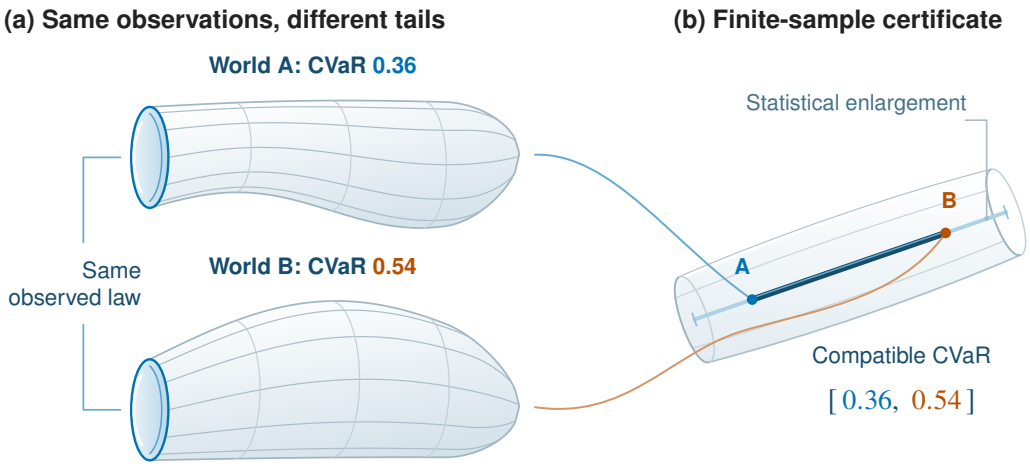


Figure 1: **Risk ambiguity and statistical uncertainty for one fixed predictor.** (a) The same observed law is compatible with hidden-outcome laws giving population CVaR 0.36 and 0.54. (b) The dark segment is the sharp population interval; pale extensions account for estimating its endpoints from a finite sample. Colors link the laws to the endpoints. Geometry is schematic. Appendix A.2 gives the numerical construction.

it is attained under the observations and the restriction. Its width measures ambiguity that persists even with unlimited historical data. Figure 1 depicts this range for one fixed predictor; the additional enlargement accounts for estimating its endpoints from a finite sample.

Our goal is to connect three questions: which risks are compatible with the evidence, how precisely their bounds can be estimated, and which additional labels most reduce unresolved ambiguity. An upper bound below a chosen loss limit certifies that criterion; separated intervals establish a predictor ordering. When these rules leave a decision unresolved, targeted outcome collection can reduce the underlying information gap.

Contributions.

1. **Sharp population identification.** For binary outcomes and bounded losses, we derive attainable CVaR endpoints and show that every intermediate risk is compatible. For each endpoint, one hidden-outcome law is extremal at every loss threshold, reducing its calculation to a scalar convex minimization (Section 3).

2. **Uniform finite-sample certification.** Paired, locally robust endpoint estimates give simultaneous bounds across candidate predictors at fixed sensitivity and tail levels. The bounds expose the costs of assessing a smaller tail, weaker observation overlap, and auxiliary-estimation error, and support comparisons and predictor selection using the same data (Section 4).

3. **Minimum-cost revelation.** Under a specified information model, ranking tail identification value per unit cost gives the exact optimal revelation allocation for a fixed predictor and threshold. A zero-threshold bound also controls optimized interval width (Section 5).

Revelation recovers existing outcomes for a fixed target population. Our information model treats audited contributions as known and retains the original sensitivity bounds separately for audited and still-hidden groups.

Synthetic experiments separate persistent identification width from decreasing endpoint error (Figure 2). Historical LendingClub data illustrate changes in population assessment, while controlled masking of known outcomes evaluates coverage and comparisons (Figure 3). Fixed and sequential audits examine how acquisition choices change estimated ambiguity (Figure 4).

Positioning. We study prediction-loss CVaR when target-population labels are selectively observed, connecting selective-label prediction and evaluation [5, 9, 17] with partial identification and sensitivity analysis [21, 23, 31, 32, 35]. CVaR optimization and distributionally robust learning [11, 12, 29] address risk under distributional ambiguity; semiparametric orthogonalization [7, 14, 27] controls auxiliary-estimation error. Our contribution combines the exact CVaR range compatible with selective observation, uniform endpoint inference, and minimum-cost allocation for fixed-threshold precision. Appendix E.3 develops this positioning.

2 Tail Risk under Selective Labels

2.1 Target population and observation process

Let $\mathcal{F}$ be the candidate predictor class and P the full-data distribution of (X, S, Y), with covariate marginal P_X. We evaluate $f \in \mathcal{F}$ using loss $\ell(f(X), Y)$, where $Y \in \{0, 1\}$. In decision settings, Y denotes $Y(a^\star)$ under one specified action held fixed across the target population and predictor comparisons. The indicator $S \in \{0, 1\}$ equals one when the outcome is observed. We observe n independent copies of $O = (X, S, Y^{\text{obs}})$, where $Y^{\text{obs}} = SY$. The observed law P_{obs} identifies P_X and

$$\pi(x) = P(S = 1 \mid X = x), \qquad p_1(x) = P(Y = 1 \mid X = x, S = 1),$$

while $p_0(x) = P(Y = 1 \mid X = x, S = 0)$ is hidden wherever $\pi(x) < 1$. Here π is the historical observation probability and p_1 the outcome probability among observed cases; both are auxiliary to the predictor f being assessed.

Assumption 1 (Bounded loss and population overlap)**.** *For every $f \in \mathcal{F}$ and $y \in \{0, 1\}$, the map $x \mapsto \ell(f(x), y)$ is measurable and takes values in $[0, 1]$, and $0 < \pi_{\min} \le \pi(X) \le 1$ almost surely.*

Population overlap means every target covariate region has a positive chance of yielding an observed label. Technical regularity and target-label conventions are in Appendix A.1. Write $z_+ = \max\{z, 0\}$ and $L_f = \ell(f(X), Y)$. At level $\alpha \in (0, 1)$, $\beta = 1 - \alpha$ is the upper-tail fraction. The threshold η is a candidate loss cutoff, and $(L_f - \eta)_+$ is the excess above it. At $\alpha = .95$, CVaR averages the worst 5% of population losses, allowing fractional mass at a tied cutoff:

$$\rho_{\alpha,P}(f) = \text{CVaR}_{\alpha,P}(L_f) = \inf_{\eta \in [0,1]} \left\{ \eta + \beta^{-1} \mathbb{E}_P[(L_f - \eta)_+] \right\}. \tag{1}$$

The expectation includes both observed and hidden cases.

2.2 Sensitivity model and certification objective

For a finite, externally specified $\Gamma \ge 1$, impose the outcome-odds restriction [31, 32, 35]

$$\begin{aligned} p_0(x) \in [\underline{p}_\Gamma(x), \overline{p}_\Gamma(x)] = [q_-(p_1(x)), q_+(p_1(x))], \qquad \pi(x) < 1, \\ q_-(p) = \frac{p}{p + \Gamma(1 - p)}, \qquad q_+(p) = \frac{\Gamma p}{1 - p + \Gamma p}. \end{aligned} \tag{2}$$

For interior probabilities, hidden-to-observed outcome odds lie between Γ^{-1} and Γ; the formulas also cover $p_1 \in \{0, 1\}$. At $\Gamma = 1$, $p_0(x) = p_1(x)$: hidden and observed outcomes have the same conditional distribution given X, called conditional ignorability. For example, $p_1(x) = .5$ and $\Gamma = 2$ allow $p_0(x) \in [1/3, 2/3]$. Larger Γ permits greater departure. The chosen Γ defines a declared sensitivity scenario, supported by domain knowledge or additional outcome information (Appendix E.1).

Let $\mathcal{P}_\Gamma(P_{\text{obs}})$ contain all full-data laws with observed marginal P_{obs} and hidden regression satisfying this restriction. Their risk endpoints are

$$\underline{\rho}_{\alpha,\Gamma}(f) = \inf_{P \in \mathcal{P}_\Gamma} \rho_{\alpha,P}(f), \qquad \overline{\rho}_{\alpha,\Gamma}(f) = \sup_{P \in \mathcal{P}_\Gamma} \rho_{\alpha,P}(f). \tag{3}$$

For fixed α and Γ, our statistical objective is an interval covering every compatible risk simultaneously for every $f \in \mathcal{F}$, with probability at least $1 - \delta_{\text{tot}}$ over the observed sample. Here $\delta_{\text{tot}} \in (0, 1)$ is the total failure budget; Section 4 accounts for sample fluctuation and auxiliary-estimation error within this budget. Thus Γ specifies the admissible hidden outcomes, while δ_{tot} specifies the statistical guarantee.

3 Sharp Population CVaR Bounds

For binary outcomes, both possible prediction losses can be computed from $f(x)$, even when the actual label is hidden. Sensitivity bounds constrain the probability assigned to these two losses. This reduces population identification to two scalar convex minimizations.

Fix a predictor f and a cutoff $\eta \in [0,1]$. Write

$$a^y_{f,\eta}(x) = (\ell(f(x),y) - \eta)_+, \qquad d_{f,\eta}(x) = a^1_{f,\eta}(x) - a^0_{f,\eta}(x).$$

Here a^y is the excess loss if the label is y, and d is its change when the label changes from zero to one. A hidden outcome probability q gives expected excess loss $(1-q)a^0 + qa^1 = a^0 + qd$. With $d_+ = (d_{f,\eta})_+$ and $d_- = (-d_{f,\eta})_+$, its largest and smallest allowed values are

$$\begin{aligned} b^U_{f,\eta} &= a^0_{f,\eta} + \overline{p}_\Gamma d_+ - \underline{p}_\Gamma d_-, \\ b^L_{f,\eta} &= a^0_{f,\eta} + \underline{p}_\Gamma d_+ - \overline{p}_\Gamma d_-. \end{aligned} \tag{4}$$

All quantities are evaluated at the same covariate value. The upper bound assigns the greatest allowed probability to the label with larger loss; the lower bound reverses the assignment. Averaging the historically observed and hidden contributions gives

$$m^\sigma_{f,\eta} = \mathbb{E}_{\mathrm{obs}}\left[S a^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1-S) b^\sigma_{f,\eta}(X)\right], \qquad \sigma \in \{L,U\},$$

where $\mathbb{E}_{\mathrm{obs}}$ integrates over the observed-data law. Adding the cutoff term forms the CVaR objectives

$$L_{f,\eta} = \eta + \beta^{-1} m^L_{f,\eta}, \qquad U_{f,\eta} = \eta + \beta^{-1} m^U_{f,\eta}.$$

Dependence on Γ is implicit.

Theorem 1 (Sharp population CVaR interval)**.** *Under Assumption 1, for every $f \in \mathcal{F}$ and finite $\Gamma \geq 1$,*

$$\underline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta \in [0,1]} L_{f,\eta}, \qquad \overline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta \in [0,1]} U_{f,\eta}.$$

Both endpoints and every intermediate value are attained by compatible full-data laws. Consequently, the set of risks $\{\rho_{\alpha,P}(f) : P \in \mathcal{P}_\Gamma(P_{\mathrm{obs}})\}$ is exactly $[\underline{\rho}_{\alpha,\Gamma}(f), \overline{\rho}_{\alpha,\Gamma}(f)]$.

The key mechanism is that thresholding preserves the ordering of the two label-specific losses. For each endpoint, the same hidden-outcome distribution is therefore extremal at every cutoff, including the optimizing cutoff. Appendices A.3–A.4 give the construction and sharpness proof, including discrete losses and ties.

The identification width

$$I_\Gamma(f) = \overline{\rho}_{\alpha,\Gamma}(f) - \underline{\rho}_{\alpha,\Gamma}(f)$$

is the unresolved range of population risk even when the observed distribution is known exactly. Sharpness means that narrowing this range requires additional information or stronger assumptions. Increasing Γ expands the compatible class and can only widen the interval; at $\Gamma = 1$, the population risk is identified under conditional ignorability.

For a chosen acceptable risk level r, $\overline{\rho}_{\alpha,\Gamma}(f) \leq r$ certifies that every compatible risk meets the limit. If $\overline{\rho}_{\alpha,\Gamma}(f) < \underline{\rho}_{\alpha,\Gamma}(g)$, the ordering holds under every compatible law. Precision and risk level are distinct: a narrow interval can still lie above the acceptable limit. Section 4 adds sampling uncertainty to these decisions.

4 Estimating and Using Tail-Risk Certificates

The observed distribution must be estimated from a finite sample. We control endpoint error uniformly across the candidate class so that the same certificate event supports comparison and selection after examining the data.

4.1 Cross-fitted endpoint estimation

The auxiliary functions $p = p_1$ and π describe the outcome among historically observed cases and the probability of historical observation. They are fitted to estimate the risk of f. Substituting an estimated p into the sensitivity bounds introduces estimation bias. We correct its leading effect using observed outcome residuals.

Write $\dot{b}^\sigma = \partial b^\sigma / \partial p$, for $\sigma \in \{L, U\}$. The locally robust score combines observed excess loss, the sensitivity-bounded hidden contribution, and the residual correction:

$$\begin{aligned}\phi^\sigma_{f,\eta}(O; p, \pi) = S a^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1 - S) b^\sigma_{f,\eta}(X; p(X)) \\ + S \frac{1 - \pi(X)}{\pi(X)} \dot{b}^\sigma_{f,\eta}(X; p(X)) \{Y^{\mathrm{obs}} - p(X)\}.\end{aligned} \tag{5}$$

At the true auxiliary functions, its expectation equals $m^\sigma_{f,\eta}$. The correction leaves quadratic and product error terms, quantified in Section 4.2 and Appendix B.1.

Split the sample into K balanced folds. Fit p, π outside each evaluation fold, then clip their predictions to $[0, 1]$ and $[\pi_{\min}/2, 1]$, respectively. Thus each score uses auxiliary fits trained without its evaluation observations. Let $n_{\min}$ be the smallest fold size and $k(i)$ the fold containing observation i. Define

$$\widehat{\rho}^\sigma(f) = \inf_{\eta \in [0,1]} \left\{ \eta + \frac{1}{n\beta} \sum_{i=1}^{n} \phi^\sigma_{f,\eta}\left(O_i; \widehat{p}^{(-k(i))}, \widehat{\pi}^{(-k(i))}\right) \right\}. \tag{6}$$

The endpoints share auxiliary fits and minimize their cutoffs separately. For each fixed f and realized auxiliary fits, exact cutoff minimization checks $0, 1$, and $\ell(f(X_i), y)$ for every $i = 1, \ldots, n$ and $y \in \{0, 1\}$ (Proposition B.5).

4.2 Uniform certificates and decision guarantees

Write $\rho^L = \underline{\rho}_{\alpha,\Gamma}$ and $\rho^U = \overline{\rho}_{\alpha,\Gamma}$.

Theorem 2 (Uniform statistical certification)**.** *Fix $\alpha \in (0, 1)$, $1 \le \Gamma < \infty$, $\delta \in (0, 1)$, and $\delta_\nu \in [0, 1 - \delta)$. Assume the bounded-loss, overlap, and sampling conditions of Section 2 and the cross-fitting and measurability conventions of Appendix B. Suppose that, with probability at least $1 - \delta_\nu$, the clipped auxiliary fits satisfy jointly*

$$\max_k \|\widehat{p}^{(-k)} - p\|_{L^2(P_X)} \le r_p, \qquad \max_k \|\widehat{\pi}^{(-k)} - \pi\|_{L^2(P_X)} \le r_\pi.$$

For the calibrated radius $r_n(\delta)$ in (B.15), with probability at least $1 - \delta - \delta_\nu$,

$$\sup_{f \in \mathcal{F}} \max_{\sigma \in \{L,U\}} |\widehat{\rho}^\sigma(f) - \rho^\sigma(f)| \le r_n(\delta). \tag{7}$$

Allocate $\delta + \delta_\nu \le \delta_{\mathrm{tot}}$ to obtain the confidence level specified in Section 2. Here δ covers sample fluctuation and δ_ν covers failure of the joint auxiliary-error bounds. The $L^2(P_X)$ bounds measure prediction error across the target covariate distribution, including regions with few observed outcomes. Computing the calibrated radius requires valid bounds on these errors and on score complexity (Appendix B.3).

The radius separates sample fluctuation from auxiliary-estimation bias. Smaller tail mass β amplifies both; weak observation overlap and larger sensitivity levels also enlarge the bound. The remaining bias contains a squared outcome-regression error r_p^2 and the product $r_p r_\pi$. At $\Gamma = 1$, the squared-error coefficient vanishes. For $\Gamma > 1$, accurate outcome regression remains important even when the observation model is known. Cross-fitting supplies the conditional independence used for concentration; the auxiliary-error bounds are established separately.

Outer certificates and decisions. Write $r_n = r_n(\delta)$. On the same event, simultaneously for all predictors,

$$[\underline{\rho}_{\alpha,\Gamma}(f), \overline{\rho}_{\alpha,\Gamma}(f)] \subseteq \widehat{C}_{n,\Gamma}(f) := [\max\{0, \widehat{\rho}^L(f) - r_n\}, \min\{1, \widehat{\rho}^U(f) + r_n\}]. \tag{8}$$

The certificate's width lies between $I_\Gamma(f)$ and $I_\Gamma(f) + 4r_n$. Its upper limit can certify a chosen risk limit. If $\widehat{\rho}^U(f) + r_n < \widehat{\rho}^L(g) - r_n$, then f has lower CVaR than g under every compatible law. Minimizing $\widehat{\rho}^U$ targets the smallest compatible upper risk in $\mathcal{F}$; Appendix B.4 gives a $2r_n$ selection-error bound with an additional term for optimization tolerance.

These decisions retain their guarantee after predictor selection within $\mathcal{F}$. As sampling error shrinks, the identification width remains. Additional labels can reduce that remaining ambiguity.

5 Resolving Tail Ambiguity through Revelation

5.1 Tail identification value

Fix f, Γ, and a cutoff $\eta \in [0, 1]$. Throughout this section, expectations are over the target covariate distribution P_X. An audit reveals existing outcomes: its indicator $A \leq 1 - S$ and intensity $\lambda(x) = P(A = 1 \mid X = x, S = 0)$ specify the share of originally hidden outcomes recovered at x.

We hold the historical observed law and the audited population contribution fixed; the remaining hidden outcomes retain the specified sensitivity bounds. Formally, let $p_A(x) = P(Y = 1 \mid X = x, A = 1)$ and $q(x) = P(Y = 1 \mid X = x, S = 0, A = 0)$. We impose the original interval $[\underline{p}_\Gamma(x), \overline{p}_\Gamma(x)]$ separately on these audited and still-hidden outcome probabilities wherever their component has positive mass; q is otherwise free within it. This defines the retained-sensitivity information model (Appendix C.1). Identification treats the audited population contribution as known; finite audited samples require estimation of that contribution.

The retained lower and upper objectives, $L_{f,\eta}(\lambda)$ and $U_{f,\eta}(\lambda)$, share the known contribution and differ only on the remaining hidden mass (Appendix C.2). Write $s_\Gamma = \overline{p}_\Gamma - \underline{p}_\Gamma$. Their gap is

$$\begin{aligned} \tau_{f,\eta}(x) &:= \frac{1 - \pi(x)}{\beta} s_\Gamma(x) |d_{f,\eta}(x)|, \\ W_{f,\eta}(\lambda) &:= U_{f,\eta}(\lambda) - L_{f,\eta}(\lambda) \\ &= \mathbb{E}[(1 - \lambda)\tau_{f,\eta}] = W_{f,\eta}(0) - \mathbb{E}[\lambda \tau_{f,\eta}]. \end{aligned} \tag{9}$$

The tail identification value (TIV) density $\tau_{f,\eta}$ combines three quantities: the population share with hidden outcomes, the allowed range of their outcome probability, and the change in excess loss between the two labels. It measures ambiguity per unit target-population mass. Revealing a fraction λ removes exactly $\mathbb{E}[\lambda \tau_{f,\eta}]$, a value calculable before seeing the audited labels. When both labels produce the same excess loss, this value is zero even if the outcome is uncertain.

5.2 Minimum-cost revelation and certificate precision

Let $c(x) \geq 0$ be an integrable cost density per unit target-population mass, so allocation λ costs $\mathbb{E}[c\lambda]$. If one already-hidden label costs $\kappa(x)$, then $c = (1 - \pi)\kappa$. For a target gap $\varepsilon \geq 0$, the minimum cost is

$$\mathrm{Cert}_{f,\eta}(\varepsilon) := \inf_{0 \leq \lambda \leq 1} \{\mathbb{E}[c\lambda] : W_{f,\eta}(\lambda) \leq \varepsilon\}. \tag{10}$$

Theorem 3 (Minimum-cost fixed-threshold revelation)**.** *Under the retained-sensitivity model, this infimum is attained. An optimum reveals zero-cost positive-TIV locations first and, on $c > 0$, satisfies*

$$\lambda^\star(x) = \begin{cases} 1, & \tau_{f,\eta}(x)/c(x) > t, \\ 0, & \tau_{f,\eta}(x)/c(x) < t, \end{cases}$$

with fractional revelation on ties, for some $t \in [0, \infty]$.

Thus acquisition follows decreasing ambiguity reduction per unit cost until the target is reached, using a partial terminal tie group when needed. This is the fractional-knapsack rule [10]. Appendices C.3 and C.5 give the proof, boundary cases, and finite-cell computation. Among already-hidden candidates with positive label cost, its operational score is

$$\frac{\tau_{f,\eta}(x)}{c(x)} = \frac{s_\Gamma(x)|d_{f,\eta}(x)|}{\beta\kappa(x)}.$$

The hidden-population fraction cancels because the candidate is already known to be hidden.

The retained endpoints minimize their cutoffs separately. Their optimized width satisfies

$$I^{\mathrm{ret}}_{\Gamma,\lambda}(f) := \min_{\eta\in[0,1]} U_{f,\eta}(\lambda) - \min_{\eta\in[0,1]} L_{f,\eta}(\lambda) \le W_{f,0}(\lambda). \tag{11}$$

Theorem 3 at $\eta = 0$ therefore minimizes cost for the sufficient requirement $W_{f,0}(\lambda) \le \varepsilon$, which guarantees optimized width at most ε. Appendix C.4 proves this bound and formulates direct optimization of the optimized interval width.

Our sequential audit procedure retrains the predictor after each batch, uses the current upper-endpoint minimizing cutoff in the retained plug-in objective, and refreshes TIV scores. Its historical outcome fit remains fixed, preserving the original sensitivity bounds. Audited labels enter through their observed losses. Candidates are ranked by estimated $s_\Gamma|d_{f,\eta}|/(\beta\kappa)$, a batch is revealed, and the procedure repeats. Figure 4 evaluates the fixed and sequential protocols; Appendix D.6 specifies fitting and evaluation splits.

6 Empirical Evaluation

We evaluate three questions: how accurately endpoints can be estimated, how selective observation changes model assessment, and which additional labels reduce unresolved ambiguity. Simulated populations provide exact identification targets. Historical applications test the consequences of changing the evaluation population. Controlled masking of known outcomes permits direct checks of estimated risks and audit performance. Appendix D provides the complete methods, results, and figure provenance.

6.1 Identification versus statistical uncertainty

We separate statistical uncertainty from population ambiguity in finite-cell populations with Brier loss $\ell(f(x), y) = (f(x) - y)^2$, $\Gamma = 2$, and $\alpha = .75$. An oracle certificate uses the true outcome probability p_1 among historically observed cases and a finite-cell Hoeffding radius. On the same sampled datasets, two-fold cross-fitting learns the nuisance functions; root-mean-square error (RMSE) assesses the resulting endpoint estimates. Each sample size uses 60 simulated populations and 80 datasets per population.

As n increases from 10^3 to 10^5, mean oracle outer width falls from .614 to .147, while mean exact population width remains .0949 (Figure 2a). The 95% oracle certificate contains the entire sharp interval in every replication; its simultaneous protection across thresholds is conservative. Upper- and lower-endpoint RMSE decrease from .0361 and .0425 to .0027 and .0023 (panel b). Larger samples sharpen estimation of the interval, while the original observation process leaves its identification width intact. The learned-nuisance experiment measures endpoint accuracy; coverage for learned fits requires the calibration conditions of Theorem 2. Appendix D.2 also reports the separate learning-bound scaling diagnostic.

6.2 Historical applicant-level sensitivity analysis

The 2012 LendingClub cohort contains 390,644 applications, including 53,367 mature issued loans with known repayment outcomes [13]. The applicant target is default under loan issuance; this outcome is unavailable for rejected applicants. The cohort includes 180,903 applications with recorded credit-score values below 660, the 2012 FICO minimum [19], and no issued loans in that group. We therefore specify an issued-loan outcome regression extended to all recorded applicant covariates. Its predictions supply the reference probabilities in (5), together with the observed-residual correction. The historical results are residual-adjusted sensitivity estimates under this regression extension. Appendix D.3 documents unsupported regions and differences in how source fields were measured.

We compare logistic regression, default-weighted logistic regression, and gradient boosting with five-fold cross-fitting. Default examples receive weight 2 in the weighted model. Standard classification objectives represent existing prediction pipelines whose population tail risks must be assessed. Brier loss is bounded and makes both possible label losses explicit. We assess tail levels $\alpha \in \{.90, .95\}$.

At $\alpha = .95$, accepted-only CVaR favors the default-weighted model (.661). At $\Gamma = 1$, its estimated applicant risk is .813, compared with .733 for boosting and .805 for logistic regression: two of three

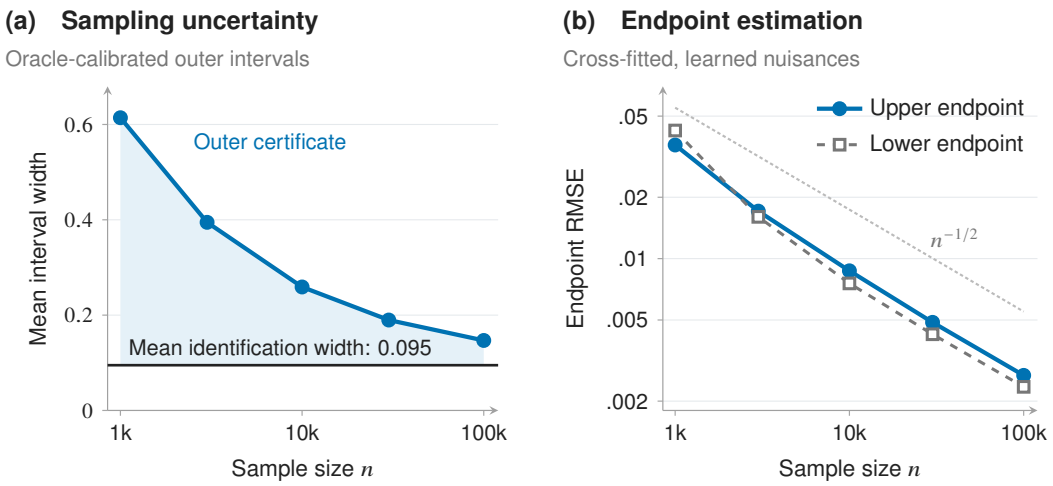


Figure 2: **Statistical uncertainty contracts around an identification floor.** (a) Mean oracle-calibrated outer width and mean exact population width; shading shows their difference. (b) Cross-fitted endpoint RMSE with learned nuisances; the faint line is an $n^{-1/2}$ guide. Each sample size uses 60 simulated populations and 80 datasets per population. The oracle coverage target is the entire sharp interval. Appendix D.2 gives the design and complete results.

pairwise orderings reverse. Figure 3a gives the intervals at $\Gamma = 2$; Appendix D.3 reports complete sensitivity curves and approximate breakdown values.

The constant forecast $f \equiv 1/2$ has loss $1/4$ under either label. Its mean Brier loss and every CVaR therefore equal $1/4$, while identification width and TIV are zero. Its tail loss is below that of the reported fitted models. Table 6 shows the competing mean-loss advantage of the audited boosting models, whose mean Brier losses are near .135. The reference separates average prediction quality, extreme-tail loss, and certainty about that loss.

6.3 Validation with known repayment outcomes

We impose 20 outcome-dependent masks on the mature issued loans, with sensitivity $\Gamma_0 = 3$ and approximately 45% of labels observed. This study targets the issued-loan reference population. Predictors are trained on observed labels using five-fold cross-fitting, and their full-outcome CVaRs provide reference risks.

At Γ_0, plug-in intervals—the estimated endpoint intervals before statistical widening—contain reference CVaR in 11–14 of 20 masks per predictor and tail level. Enlargement using variability across five fold-specific objectives raises coverage to 19–20 (Figure 3b). Coverage here targets the full-outcome reference CVaR of each fitted predictor. Fold enlargement provides an empirical calibration diagnostic. Plug-in coverage is zero at every tested $\Gamma \leq 2$.

Three unordered predictor pairs in each of 20 masks give 60 comparisons per tail level. At $\Gamma = 3$, $\alpha = .95$, strict interval separation resolves 37 pairs correctly and leaves 23 unresolved. At $\Gamma = 1$, the rule declares all 120 orderings across both tail levels and makes 16 errors. Allowing the masking sensitivity changes both the inferred ordering and whether separation supports a decision. Appendix D.4 gives the mechanism, enlargement rule, and complete results.

6.4 Fixed and sequential audits

In 120 synthetic finite-cell populations with heterogeneous costs, TIV per cost matches the exact fractional-knapsack frontier for fixed predictor and threshold (Appendix D.5). We then audit the masked issued-loan population using boosting, $\alpha = .95$, $\Gamma = 3$, and unit cost per hidden label.

The fixed protocol freezes the initial predictor, historical nuisance estimates, and upper-endpoint threshold. Newly audited outcomes become known contributions, while the original sensitivity

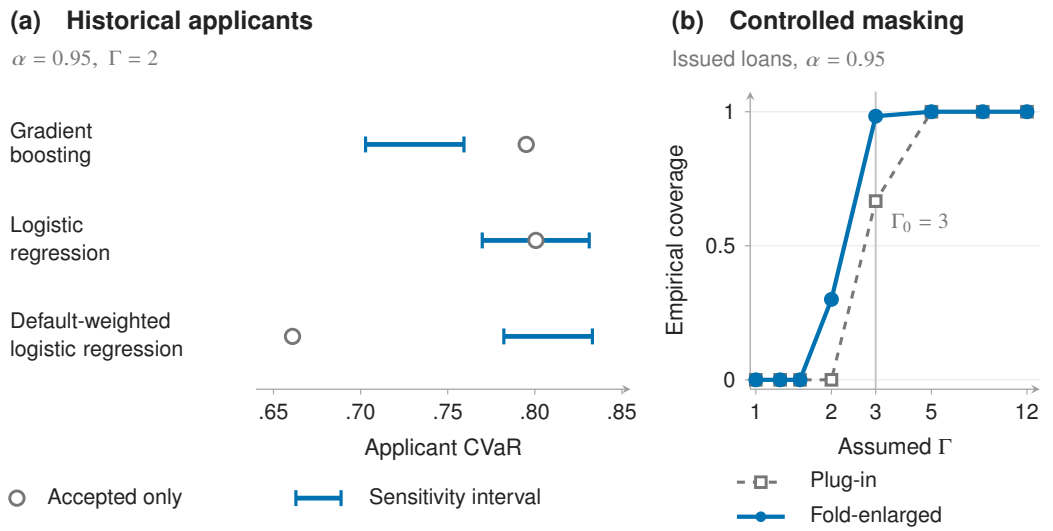


Figure 3: **Observation and sensitivity change population assessment.** (a) Accepted-only CVaR (hollow points) and estimated applicant sensitivity endpoints (blue segments) at $\alpha = .95$, $\Gamma = 2$, under the regression extension of Appendix D.3. Segments precede statistical enlargement. (b) Coverage of full-outcome reference CVaR under controlled masking, pooled across three predictors and 20 shared masks at $\alpha = .95$. The guide marks the masking mechanism's sensitivity $\Gamma_0 = 3$; fold enlargement is empirical. Appendix D.4 provides the complete results.

bounds remain on unaudited outcomes. The sequential protocol retrains the predictor and reoptimizes thresholds after each batch; retained evaluation keeps the historical outcome fit fixed, and a reserved split measures predictive loss. The lowest-propensity baseline selects cases with the lowest estimated label availability; it uses historical availability in the fixed protocol and refitted current known-label availability in the sequential protocol.

At the recorded budget of approximately 16%, TIV removes a median 61% of the initial estimated fixed-threshold gap, compared with 16% for random acquisition (Figure 4a). This concentration of removable ambiguity explains the gain from targeting. For Brier loss, predictions near .5 have similar losses under either label, so uncertainty sampling can favor cases with small excess-loss contrasts and little identification value.

With sequential retraining, median estimated retained width falls from .097 to .035 under TIV (panel b). This width describes the current predictor; it can increase when retraining changes the assessed losses, as the other trajectories illustrate (Table 5). Terminal median held-out CVaR is .8541 for TIV and .8682 for random, while mean Brier loss is .1362 and .1347, respectively (Table 6). Relative to random acquisition, these runs show improved tail performance and precision alongside a small mean-loss trade-off. Appendix D.6 gives all policies and both protocols.

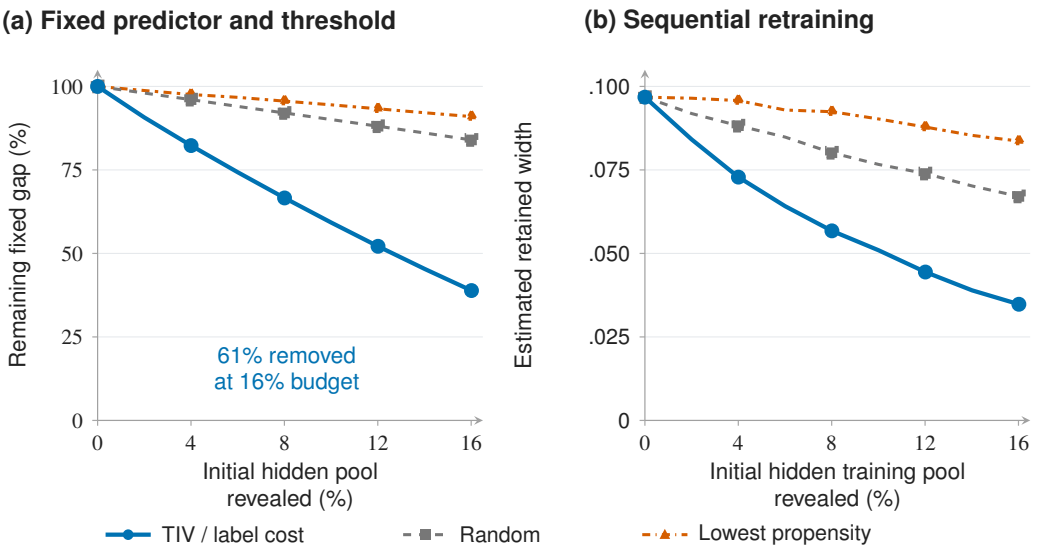


Figure 4: **Targeted audits reduce estimated ambiguity at equal within-protocol budgets.** (a) Median remaining fixed-threshold gap, normalized within each mask by its initial estimated gap. (b) Median estimated retained-sensitivity width for the predictor refitted after each audit batch. Both use boosting, $\alpha = .95$, $\Gamma = 3$, unit hidden-label cost, and 20 masks. Lines join eight recorded batches. Appendix D.6 specifies both protocols. Budgets are median fractions of the initially hidden cases in each eligible pool: the original pool in (a) and the 80% training pool in (b).

7 Limitations

The sensitivity range must be justified for the target population and covariate resolution. Statistical certificates additionally require overlap, controlled score complexity, and valid nuisance-error bounds. Small tail mass and scarce observed labels can make the bounds conservative. For $\Gamma > 1$, outcome-regression error contributes quadratically even with a correct observation model. Choosing Γ from data or examining multiple sensitivity levels requires corresponding joint error control (Appendix E.1).

The historical applicant analysis depends on regression extrapolation into unsupported regions and on differently measured source fields. Controlled masking checks known issued-loan outcomes. Repayment under an untaken lending action requires a causal or experimental model beyond passive recovery of existing outcomes.

The revelation optimum fixes the predictor and threshold and assumes componentwise retained sensitivity bounds. Known outcome-independent audit randomization can identify more, and direct minimization of the reoptimized interval can favor a different allocation. Finite audits must estimate their newly known contributions. Learned acquisition scores, retraining, and data-dependent stopping require uncertainty control across the resulting audit history; the sequential trajectories provide empirical performance evidence. Extensions to richer outcomes, unbounded losses, and other application domains require additional modeling and validation.

8 Conclusion

Selective observation leaves a range of population tail risks compatible with the available outcomes. Sharp CVaR endpoints characterize that range, and calibrated uniform outer certificates support risk limits, model comparisons, and conservative selection within a candidate class. Under retained sensitivity, TIV quantifies the ambiguity removed by revelation for a fixed predictor and threshold. Ranking this value per cost gives the optimal population allocation for that objective. These outputs connect a specified risk decision to the observations and assumptions required to support it, and identify where further outcome collection can improve precision.

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Guide to the appendices

Appendix A gives compatible laws, examples, and identification proofs; Appendix B gives score bias, uniform estimation, calibration, and the learning lower bound; Appendix C gives retained-information revelation, its cost frontier, and computation. Appendix D records scientific protocols and complete experimental results; Appendix E treats sensitivity calibration, extensions, and related approaches. Implementation provenance accompanies the source in `reproducibility/`.

Throughout, $\beta = 1 - \alpha$, $\Gamma < \infty$, and the target population and outcome are fixed. $I_\Gamma(f)$ is identification width; $\widehat{C}_{n,\Gamma}(f)$ is a statistical outer certificate. After revelation, $W_{f,\eta}(\lambda)$ is the gap at a common threshold, whereas $I^{\mathrm{ret}}_{\Gamma,\lambda}(f)$ optimizes the two endpoints separately. Auxiliary results and equations use appendix-local numbering.

A Population Identification and Sensitivity

A.1 Compatible full-data laws and sensitivity bounds

The target outcome and population are fixed across predictors and compatible laws. In a decision problem, $Y = Y(a^\star)$ is the outcome under a specified action (e.g., default under loan issuance), and S records its availability [15–17]. Using an observed outcome requires consistency with that action and outcome definition. Comparing prediction rules holds $a^\star$ and P_X fixed; changing the action, recovering an outcome under an untaken action, or transferring to a new population requires an additional causal or transport argument. The sensitivity model supplies neither exchangeability nor causal identification. Passive revelation recovers an already-realized target outcome, as in record linkage or controlled masking. Thus mature issued loans supply the masking target, whereas repayment under issuance remains a potential outcome for rejected applicants in the historical analysis.

Let $\mathcal{X}$ be standard Borel, and let P_{obs} be the law of $O = (X, S, Y^{\mathrm{obs}})$ with $Y^{\mathrm{obs}} = SY$. Use measurable versions

$$\pi(x) = P_{\mathrm{obs}}(S = 1 \mid X = x), \qquad p_1(x) = P_{\mathrm{obs}}(Y^{\mathrm{obs}} = 1 \mid X = x, S = 1),$$

and losses $z^y_f(x) = \ell(f(x), y) \in [0, 1]$. Assumption 1 gives $\pi \geq \pi_{\min} > 0$, so $\pi\, dP_X$ and P_X have the same null sets and p_1 is identified P_X-almost surely. All unqualified almost-sure statements use P_X. Write $H = \{\pi < 1\}$. For a measurable $q : \mathcal{X} \to [0, 1]$ and its Bernoulli kernel K_q, define

$$\begin{aligned} P^q(dx, \{1\}, dy) &= P_X(dx)\pi(x)K_{p_1}(x, dy), \\ P^q(dx, \{0\}, dy) &= P_X(dx)[1 - \pi(x)]K_q(x, dy). \end{aligned} \tag{A.1}$$

Set $q = p_1$ on H^c, where its value carries zero hidden mass.

Lemma A.1 (Kernel representation of compatible laws)**.** *Every full-data law inducing P_{obs} has the form (A.1), with $q(x) = P(Y = 1 \mid X = x, S = 0)$ on H. Conversely, every measurable $q : \mathcal{X} \to [0, 1]$ in (A.1) induces P_{obs}. With $q = p_1$ on H^c, these kernels identify the compatible laws up to P_X-null sets.*

Proof. Standard Borel disintegration fixes the (X, S) law and selected-label kernel, leaving one binary probability on $S = 0$. Conversely, the two displayed measures sum to one and have the prescribed pushforward under $(x, s, y) \mapsto (x, s, sy)$. Equal full-data laws have equal hidden kernels $(1 - \pi)dP_X$-almost surely, equivalently P_X-almost surely on H; the canonical assignment settles H^c. □

For finite $\Gamma \geq 1$, the odds restriction is equivalent, at interior p_1, to the bounds $\underline{p}_\Gamma = q_-(p_1; \Gamma)$ and $\overline{p}_\Gamma = q_+(p_1; \Gamma)$, where

$$q_-(p; \Gamma) = \frac{p}{p + \Gamma(1 - p)}, \qquad q_+(p; \Gamma) = \frac{\Gamma p}{1 - p + \Gamma p}. \tag{A.2}$$

Their denominators are at least one. Thus $p_1 = 0$ forces $q = 0$ and $p_1 = 1$ forces $q = 1$ at every finite Γ; these substantive boundary restrictions merit scrutiny when apparent certainty comes from

a fitted model. For interior p_1, the endpoints tend to zero and one as $\Gamma \to \infty$. Let $\mathcal{Q}_\Gamma$ contain measurable kernels within these bounds on H, canonically assigned on H^c. Then

$$\mathcal{P}_\Gamma(P_{\text{obs}}) = \{P^q : q \in \mathcal{Q}_\Gamma\}. \tag{A.3}$$

It is nonempty since $\underline{p}_\Gamma \le p_1 \le \overline{p}_\Gamma$; at $\Gamma = 1$ it is the single law satisfying $Y \perp S \mid X$.

For $\beta = 1 - \alpha \in (0, 1)$, the CVaR representation [29] reduces to

$$\rho_{\alpha,P}(f) = \inf_{\eta \in \mathbb{R}} \{\eta + \beta^{-1}\mathbb{E}_P(L_f - \eta)_+\} = \min_{\eta \in [0,1]} \{\eta + \beta^{-1}\mathbb{E}_P(L_f - \eta)_+\}. \tag{A.4}$$

For $\eta < 0$, the objective is $\beta^{-1}\mathbb{E}_P L_f + (1 - \beta^{-1})\eta$ and exceeds its value at zero; for $\eta > 1$, it equals $\eta > 1$. Continuity on $[0, 1]$ gives attainment, including distributions with atoms.

A.2 Observational equivalence and motivating constructions

Theorem A.2 (Observational equivalence with reversed CVaR rankings)**.** *For every $\alpha \in (0, 1)$, there exist an observed law satisfying strict overlap, two predictors f, g with identical selected-loss distributions, and two full-data laws $P^\star, Q^\star$ inducing that observed law such that*

$$\rho_{\alpha,P^\star}(f) < \rho_{\alpha,P^\star}(g), \qquad \rho_{\alpha,Q^\star}(f) > \rho_{\alpha,Q^\star}(g).$$

Proof. Take constant X, $0 < \pi < \min\{1, 2\beta\}$, $S \sim \text{Bernoulli}(\pi)$, and $Y \mid S = 1 \sim \text{Bernoulli}(1/2)$. Under zero-one loss, the constant predictors $f = 0, g = 1$ have the same selected-loss law. Set $Y = 0$ on $S = 0$ under $P^\star$ and $Y = 1$ there under $Q^\star$. Both yield observed masses $\pi/2, \pi/2, 1 - \pi$ on $(S, Y^{\text{obs}}) = (1, 1), (1, 0), (0, 0)$ and satisfy overlap with $\pi_{\min} = \pi$.

For $B \sim \text{Bernoulli}(r)$, the threshold objective is $r/\beta + \eta(1 - r/\beta)$ on $[0, 1]$, hence

$$\text{CVaR}_\alpha(B) = \min\{r/\beta, 1\}. \tag{A.5}$$

Under $P^\star$, the loss probabilities are $\pi/2$ and $1 - \pi/2$, so $\rho_{\alpha,P^\star}(f) = \pi/(2\beta) < \min\{(1 - \pi/2)/\beta, 1\} = \rho_{\alpha,P^\star}(g)$. Under $Q^\star$ they exchange. □

This construction concerns the unrestricted compatible class: its hidden probabilities zero and one lie outside the finite-Γ interval $[1/(1 + \Gamma), \Gamma/(1 + \Gamma)]$ when $p_1 = 1/2$.

Figure 1. For squared loss, $f = 1/5$, $\pi = 1/2$, $p_1 = 1/3$, $\Gamma = 2$, and $\alpha = 1/2$, the loss is $1/25 + (3/5)B$ with $B \sim \text{Bernoulli}(1/6 + q/2)$ and $q \in [1/5, 1/2]$. Translation and positive scaling of (A.4), followed by (A.5), give

$$\begin{aligned} \rho_{1/2,P^q}(f) &= \frac{1}{25} + \frac{6}{5}\left(\frac{1}{6} + \frac{q}{2}\right), \\ \rho_{1/2,P^{1/5}}(f) = \frac{9}{25} = 0.36, &\quad \rho_{1/2,P^{1/2}}(f) = \frac{27}{50} = 0.54. \end{aligned} \tag{A.6}$$

These endpoints describe the risk of one fixed predictor under the same observed law.

A.3 Conditional excess-loss envelopes and common extremal laws

Write $a^y_{f,\eta} = (z^y_f - \eta)_+$, $d_{f,\eta} = a^1_{f,\eta} - a^0_{f,\eta}$, $d_+ = (d_{f,\eta})_+$, and $d_- = (-d_{f,\eta})_+$. The hidden conditional excess loss is $a^0 + qd$, with envelopes

$$\begin{aligned} b^U_{f,\eta} &= a^0_{f,\eta} + \overline{p}_\Gamma d_+ - \underline{p}_\Gamma d_-, \\ b^L_{f,\eta} &= a^0_{f,\eta} + \underline{p}_\Gamma d_+ - \overline{p}_\Gamma d_-. \end{aligned} \tag{A.7}$$

The observable objectives are

$$\begin{aligned} m^\sigma_{f,\eta} &= \mathbb{E}_{\text{obs}}[S a^{Y^{\text{obs}}}_{f,\eta}(X) + (1 - S) b^\sigma_{f,\eta}(X)], \quad \sigma \in \{L, U\}, \\ L_{f,\eta} &= \eta + \beta^{-1} m^L_{f,\eta}, \qquad U_{f,\eta} = \eta + \beta^{-1} m^U_{f,\eta}. \end{aligned} \tag{A.8}$$

Lemma A.3 (Common extremal kernels and threshold regularity)**.** *On H, define*

$$\begin{aligned} q_f^U(x) &= \overline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) \geq z_f^0(x)\} + \underline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) < z_f^0(x)\}, \\ q_f^L(x) &= \underline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) \geq z_f^0(x)\} + \overline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) < z_f^0(x)\}, \end{aligned} \tag{A.9}$$

and set both kernels to p_1 on H^c. These kernels belong to $\mathcal{Q}_\Gamma$. For every $q \in \mathcal{Q}_\Gamma$ and every $\eta \in [0,1]$,

$$m_{f,\eta}^L \leq \mathbb{E}_{P^q}[(L_f - \eta)_+] \leq m_{f,\eta}^U, \qquad \mathbb{E}_{P^{q_f^\sigma}}[(L_f - \eta)_+] = m_{f,\eta}^\sigma. \tag{A.10}$$

The objectives $L_{f,\eta}, U_{f,\eta}$ are convex and $(1+\beta^{-1})$-Lipschitz in η, and attain their minima on $[0,1]$.

Proof. The kernels are measurable and satisfy the bounds. Since $z \mapsto (z-\eta)_+$ is nondecreasing, thresholding preserves the ordering of z_f^1, z_f^0; it can create a tie but cannot reverse the sign. Thus the same kernels extremize the affine expression $a^0 + qd$ at every threshold, including ties. Disintegration gives

$$\begin{aligned} \mathbb{E}_{P^q}(L_f - \eta)_+ = \mathbb{E}_{\mathrm{obs}}[Sa_{f,\eta}^{Y^{\mathrm{obs}}}(X)] \\ + \mathbb{E}_{P_X}[(1-\pi)(a_{f,\eta}^0 + qd_{f,\eta})], \end{aligned} \tag{A.11}$$

proving (A.10), with zero contribution from H^c. Each endpoint objective is therefore the CVaR objective of one fixed law. Convexity and the Lipschitz bound follow from the convex, 1-Lipschitz hinge; compactness gives attainment. □

A.4 Sharpness, endpoint attainment, and interval filling

Proof of Theorem 1. Put $G_f(q,\eta) = \eta + \beta^{-1}\mathbb{E}_{P^q}(L_f - \eta)_+$. Lemma A.3 gives $L_{f,\eta} = G_f(q_f^L, \eta) \leq G_f(q,\eta) \leq G_f(q_f^U,\eta) = U_{f,\eta}$ simultaneously for all thresholds. Minimizing these inequalities and choosing the respective common kernels proves

$$\underline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta\in[0,1]} L_{f,\eta}, \qquad \overline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta\in[0,1]} U_{f,\eta}. \tag{A.12}$$

Both threshold minima and both optimized endpoints are attained. In particular, the upper equality uses one kernel attaining the supremum at every threshold, without an abstract minimax interchange.

For functions with finite infima on a common domain,

$$\left|\inf_\eta F(\eta) - \inf_\eta G(\eta)\right| \leq \sup_\eta |F(\eta) - G(\eta)|, \tag{A.13}$$

because each function is bounded below by the other minus their uniform distance. Equation (A.11) and $|d_{f,\eta}| \leq 1$ therefore imply

$$|\rho_{\alpha,P^q}(f) - \rho_{\alpha,P^{\widetilde{q}}}(f)| \leq \sup_\eta |G_f(q,\eta) - G_f(\widetilde{q},\eta)| \leq \beta^{-1}\mathbb{E}[(1-\pi)|q - \widetilde{q}|].$$

The mixture $q_t = (1-t)q_f^L + tq_f^U$, $t \in [0,1]$, remains measurable and compatible, and satisfies

$$|\rho_{\alpha,P^{q_t}}(f) - \rho_{\alpha,P^{q_s}}(f)| \leq \frac{|t-s|}{\beta}\mathbb{E}[(1-\pi)|q_f^U - q_f^L|] \leq \frac{|t-s|}{\beta}. \tag{A.14}$$

This continuous path joins the two attained endpoints. The intermediate value theorem, including the singleton case, yields

$$\{\rho_{\alpha,P}(f) : P \in \mathcal{P}_\Gamma(P_{\mathrm{obs}})\} = [\underline{\rho}_{\alpha,\Gamma}(f), \overline{\rho}_{\alpha,\Gamma}(f)]. \tag{A.15}$$

□

A.5 Sensitivity analysis and population decision criteria

Proposition A.4 (Nested classes and continuous endpoint curves)**.** *If $1 \le \Gamma_1 \le \Gamma_2 < \infty$, then*

$$\mathcal{P}_{\Gamma_1}(P_{\mathrm{obs}}) \subseteq \mathcal{P}_{\Gamma_2}(P_{\mathrm{obs}}).$$

For each f, the lower endpoint is nonincreasing in Γ, the upper endpoint is nondecreasing, and the identification width $I_\Gamma(f) = \overline{\rho}_{\alpha,\Gamma}(f) - \underline{\rho}_{\alpha,\Gamma}(f)$ is nondecreasing. The interval is a singleton at $\Gamma = 1$. Moreover, for $\sigma \in \{L, U\}$, writing $\rho^L_{\alpha,\Gamma} = \underline{\rho}_{\alpha,\Gamma}$ and $\rho^U_{\alpha,\Gamma} = \overline{\rho}_{\alpha,\Gamma}$,

$$|\rho^\sigma_{\alpha,\Gamma_2}(f) - \rho^\sigma_{\alpha,\Gamma_1}(f)| \le \frac{\mathbb{E}_{P_X}(1-\pi)}{4\beta}|\Gamma_2 - \Gamma_1|. \tag{A.16}$$

In particular, both endpoint curves are continuous at every finite sensitivity level.

Proof. For $p \in [0,1]$, $\partial_\Gamma q_- = -p(1-p)/\{p+\Gamma(1-p)\}^2 \in [-1/4, 0]$ and $\partial_\Gamma q_+ = p(1-p)/\{1-p+\Gamma p\}^2 \in [0, 1/4]$. Their signs prove nesting and endpoint monotonicity; at one both maps equal p. Their magnitude bounds and the Γ-independent loss ordering in (A.9) bound the difference of the extremal kernels by $|\Gamma_2 - \Gamma_1|/4$. Apply (A.11) and (A.13) to obtain (A.16). □

Corollary A.5 (Population risk budgets, comparisons, and selection)**.** *For any risk budget r,*

$$\rho_{\alpha,P}(f) \le r \quad \textit{for every } P \in \mathcal{P}_\Gamma(P_{\mathrm{obs}}) \quad \Longleftrightarrow \quad \overline{\rho}_{\alpha,\Gamma}(f) \le r. \tag{A.17}$$

For two predictors, the sufficient separation condition

$$\overline{\rho}_{\alpha,\Gamma}(f) < \underline{\rho}_{\alpha,\Gamma}(g) \tag{A.18}$$

implies $\rho_{\alpha,P}(f) < \rho_{\alpha,P}(g)$ under every compatible law. If $\overline{\rho}_{\alpha,\Gamma}(f_\varepsilon) \le \inf_{g\in\mathcal{F}} \overline{\rho}_{\alpha,\Gamma}(g) + \varepsilon$, then every such law satisfies $\rho_{\alpha,P}(f_\varepsilon) \le \inf_{g\in\mathcal{F}} \overline{\rho}_{\alpha,\Gamma}(g) + \varepsilon$. A nonempty finite class permits $\varepsilon = 0$.

Proof. Use the attained upper endpoint for (A.17), the chain $\rho_P(f) \le \overline{\rho}(f) < \underline{\rho}(g) \le \rho_P(g)$ for comparison, and $\rho_P(f_\varepsilon) \le \overline{\rho}(f_\varepsilon)$ for selection. □

If $\underline{\rho}(f) > r$, every law violates the budget; if $\underline{\rho}(f) \le r < \overline{\rho}(f)$, the two extremal laws respectively satisfy and violate it. Thus sharpness covers all budget and equality cases.

Common-law comparisons. Define $D_{f,g}(\Gamma) = \overline{\rho}_{\alpha,\Gamma}(f) - \underline{\rho}_{\alpha,\Gamma}(g)$. Marginal extrema can use different hidden kernels, whereas the direct contrast is

$$\Delta_\Gamma(f,g) = \sup_{P\in\mathcal{P}_\Gamma(P_{\mathrm{obs}})} \{\rho_{\alpha,P}(f) - \rho_{\alpha,P}(g)\} \le D_{f,g}(\Gamma). \tag{A.19}$$

Consequently, $D_{f,g} < 0$ proves common-law dominance, but overlapping marginal intervals leave its sign unresolved by this criterion. Determining Δ_Γ requires joint optimization under one hidden kernel. The uniform event of Appendix B.4 makes the marginal comparison and selection rules valid simultaneously over the stipulated predictor class using the same observations.

First breakdown of an interval-certified comparison. Set $\Gamma_{\mathrm{break}}(f,g) = \inf\{\Gamma \ge 1 : D_{f,g}(\Gamma) \ge 0\}$, with $\inf \varnothing = \infty$. The gap is continuous and nondecreasing. If $D_{f,g}(1) \ge 0$, breakdown is one and no strict interval certificate holds. If $D_{f,g}(1) < 0$, breakdown exceeds one; when finite, continuity forces $D_{f,g}(\Gamma_{\mathrm{break}}) = 0$ and strict separation holds exactly on $[1, \Gamma_{\mathrm{break}})$. A nonzero gap at that boundary would persist across it, contradicting first failure. Infinite breakdown means separation at every finite level. Differences between two nondecreasing upper endpoints need not be monotone, so the upper-endpoint minimizer may change independently of these pairwise breakdown values.

B Statistical Certificates and Learning Complexity

This appendix proves the statistical results for a fixed tail level $\alpha \in (0,1)$ and a fixed finite sensitivity level $\Gamma \geq 1$. Write $\beta = 1-\alpha$, $p = p_1$, and $P = P_{\mathrm{obs}}$; thus Pg denotes integration of an observed-data function g against the common observed law. For brevity, set

$$\rho^L(f) = \underline{\rho}_{\alpha,\Gamma}(f), \qquad \rho^U(f) = \overline{\rho}_{\alpha,\Gamma}(f), \qquad I_\Gamma(f) = \rho^U(f) - \rho^L(f).$$

The population characterization in Appendix A gives $\rho^\sigma(f) = \inf_{\eta\in[0,1]}\{\eta + \beta^{-1} m^\sigma_{f,\eta}\}$ for $\sigma \in \{L,U\}$. All L_2 norms below are with respect to P_X.

For uniform empirical-process statements, assume the paired maps $x \mapsto (\ell(f(x),0), \ell(f(x),1))$, $f \in \mathcal{F}$, form a pointwise-separable class of measurable functions with the envelope supplied by Assumption 1. A countable pointwise-dense subclass and rational thresholds including $0, 1$ determine the suprema below, since the score transformations are continuous. Measurable, pointwise-separable versions of the fitted score classes suffice as well.

B.1 Score regularity and exact nuisance-bias expansion

Recall $a^y_{f,\eta}(x) = (\ell(f(x),y) - \eta)_+$ and $d_{f,\eta}(x) = a^1_{f,\eta}(x) - a^0_{f,\eta}(x)$. For a real number d, write $d_+ = \max(d,0)$ and $d_- = \max(-d,0)$. Define

$$H^U_p(d) = q_+(p)d_+ - q_-(p)d_-, \qquad H^L_p(d) = q_-(p)d_+ - q_+(p)d_-, \tag{B.1}$$

$$G^U_p(d) = q'_+(p)d_+ - q'_-(p)d_-, \qquad G^L_p(d) = q'_-(p)d_+ - q'_+(p)d_-. \tag{B.2}$$

Consequently, $b^\sigma_{f,\eta}(x;p) = a^0_{f,\eta}(x) + H^\sigma_p(d_{f,\eta}(x))$ and $\dot{b}^\sigma_{f,\eta}(x;p) = G^\sigma_p(d_{f,\eta}(x))$.

Lemma B.1 (Regularity of the endpoint maps)**.** *For $p \in [0,1]$, the derivatives of the sensitivity endpoints satisfy*

$$q'_+(p) = \frac{\Gamma}{\{1+(\Gamma-1)p\}^2}, \qquad q'_-(p) = \frac{\Gamma}{\{\Gamma-(\Gamma-1)p\}^2}, \tag{B.3}$$

$$q''_+(p) = -\frac{2\Gamma(\Gamma-1)}{\{1+(\Gamma-1)p\}^3}, \qquad q''_-(p) = \frac{2\Gamma(\Gamma-1)}{\{\Gamma-(\Gamma-1)p\}^3}. \tag{B.4}$$

In particular, $\Gamma^{-1} \leq q'_\pm(p) \leq \Gamma$ and $|q''_\pm(p)| \leq 2\Gamma(\Gamma-1)$. Uniformly in (f,η,x,p,σ),

$$0 \leq b^\sigma_{f,\eta}(x;p) \leq 1, \qquad |\dot{b}^\sigma_{f,\eta}(x;p)| \leq \Gamma, \qquad |\partial_p^2 b^\sigma_{f,\eta}(x;p)| \leq 2\Gamma(\Gamma-1). \tag{B.5}$$

Moreover, H^σ_p is 1-Lipschitz and G^σ_p is Γ-Lipschitz on the real line; both vanish at zero.

Proof. Differentiating $q_+(p) = \Gamma p/\{1+(\Gamma-1)p\}$ and $q_-(p) = p/\{\Gamma-(\Gamma-1)p\}$ gives (B.3)–(B.4). The denominators before taking powers lie in $[1,\Gamma]$, giving the derivative bounds. The value $a^0 + H^\sigma_p(a^1 - a^0)$ is a convex combination of $a^0, a^1 \in [0,1]$; its derivatives are bounded by $\Gamma|a^1-a^0|$ and $2\Gamma(\Gamma-1)|a^1-a^0|$, respectively. This proves (B.5). On either half-line the slopes of H^σ_p and G^σ_p lie in $[0,1]$ and $[\Gamma^{-1},\Gamma]$, respectively. Both maps are continuous at zero, so these slope bounds also prove the Lipschitz bounds across loss ties. □

For candidate nuisance functions $p_c : \mathcal{X} \to [0,1]$ and $\pi_c : \mathcal{X} \to [\pi_{\min}/2, 1]$, the score is

$$\begin{aligned}\phi^\sigma_{f,\eta}(O;p_c,\pi_c) = {} & S a^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1-S) b^\sigma_{f,\eta}(X;p_c(X)) \\ & + S\frac{1-\pi_c(X)}{\pi_c(X)} \dot{b}^\sigma_{f,\eta}(X;p_c(X))\{Y^{\mathrm{obs}} - p_c(X)\}.\end{aligned} \tag{B.6}$$

The sum of its first two terms belongs to $[0,1]$, whereas its correction has absolute value at most $2\Gamma/\pi_{\min}$. Therefore

$$\begin{aligned}-\frac{2\Gamma}{\pi_{\min}} \leq \phi^\sigma_{f,\eta}(O;p_c,\pi_c) \leq 1 + \frac{2\Gamma}{\pi_{\min}}, \\ |\phi^\sigma_{f,\eta}(O;p_c,\pi_c)| \leq B := 1 + \frac{2\Gamma}{\pi_{\min}}.\end{aligned} \tag{B.7}$$

The score can be negative; the concentration argument below explicitly allows signed classes.

Lemma B.2 (Exact conditional nuisance bias)**.** *Suppress (f, η, σ, X) inside b and its derivatives. With $e = p_c - p$, the conditional difference between the score expectation and the target excess-loss integrand is exactly*

$$(1-\pi)\{b(p_c) - b(p) - \dot{b}(p_c)e\} + \frac{\pi_c - \pi}{\pi_c}\dot{b}(p_c)e. \tag{B.8}$$

Taking P_X expectations of (B.8) *gives $P\phi^{\sigma}_{f,\eta}(\cdot; p_c, \pi_c) - m^{\sigma}_{f,\eta}$, and*

$$\begin{aligned}&\left|P\phi^{\sigma}_{f,\eta}(\cdot; p_c, \pi_c) - m^{\sigma}_{f,\eta}\right| \\ &\quad\le \Gamma(\Gamma-1)\|p_c - p\|_2^2 + \frac{2\Gamma}{\pi_{\min}}\|p_c - p\|_2\|\pi_c - \pi\|_2.\end{aligned} \tag{B.9}$$

Proof. Conditional on X, the observed excess-loss contribution has mean $\pi\{(1-p)a^0 + pa^1\}$ and $\mathbb{E}[S(Y^{\mathrm{obs}} - p_c) \mid X] = \pi(p - p_c)$. Subtracting the target integrand $\pi\{(1-p)a^0 + pa^1\} + (1-\pi)b(p)$ leaves

$$(1-\pi)\{b(p_c) - b(p)\} + \frac{\pi(1-\pi_c)}{\pi_c}\dot{b}(p_c)(p - p_c).$$

The identity $\pi(1-\pi_c)/\pi_c = (1-\pi) + (\pi - \pi_c)/\pi_c$ reduces this expression to (B.8). Taylor's theorem, expanded about p_c along the segment joining p_c and p, and Lemma B.1 give

$$|b(p_c) - b(p) - \dot{b}(p_c)(p_c - p)| \le \Gamma(\Gamma-1)(p_c - p)^2.$$

Use $0 \le 1 - \pi \le 1$ for the first term, and $|\dot{b}(p_c)/\pi_c| \le 2\Gamma/\pi_{\min}$ followed by Cauchy–Schwarz for the second term. This yields (B.9) uniformly over all score indices. □

The score is unbiased at the true nuisances and has $O(t^2)$ bias along admissible nuisance paths with L_2 perturbations of order t. The sign of $d_{f,\eta}(x)$ is independent of the nuisance argument, so endpoint selection adds no remainder. At $\Gamma = 1$ the curvature term vanishes; for $\Gamma > 1$, knowing π alone generally leaves quadratic error in p. Thus the result is an orthogonal expansion, not generic double robustness. Its overlap premise and clipping ranges are substantive: clipping cannot supply labels or identification where overlap fails.

B.2 Cross-fitting and uniform empirical-process control

Let $I_1, \dots, I_K$ be a deterministic partition of the observation indices, with $n_k = |I_k| \ge 1$, $n_{\min} = \min_k n_k$, and $w_k = n_k/n$. A random partition independent of the observations is also allowed by conditioning on that partition. Write D_{-k} for the training observations outside I_k and $P_{n,k}$ for the empirical measure on I_k. The clipped fits $\widehat{p}^{(-k)}, \widehat{\pi}^{(-k)}$ are measurable functions of D_{-k}. Any independent fitting randomization can be included in this conditioning. Define

$$\phi^{\sigma,(k)}_{f,\eta} = \phi^{\sigma}_{f,\eta}(\cdot; \widehat{p}^{(-k)}, \widehat{\pi}^{(-k)}), \qquad \Phi^{\sigma}_k = \{\phi^{\sigma,(k)}_{f,\eta} : f \in \mathcal{F},\ \eta \in [0,1]\}.$$

The empirical Rademacher complexities used here are

$$\begin{aligned}\widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}} &= \mathbb{E}_{\varepsilon}\left[\sup_{f,\eta}\left|\frac{1}{n_k}\sum_{i\in I_k}\varepsilon_i\phi^{\sigma,(k)}_{f,\eta}(O_i)\right| \,\middle|\, D_n\right], \\ \widehat{\mathfrak{R}}^{\sigma}_{n,\mathrm{abs}} &= \sum_{k=1}^{K} w_k \widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}}.\end{aligned} \tag{B.10}$$

Here D_n includes the observations, partition, and fitting seeds. The expectation averages only independent uniform signs in $\{-1, 1\}$; the fitted functions and evaluation observations are fixed.

Lemma B.3 (Empirical-complexity bound for a signed class)**.** *Let $Z_1, \dots, Z_m$ be independent with law Q and empirical measure Q_m, and let $\mathcal{H}$ be a fixed pointwise-separable class with $|h| \le B$. For $0 < \xi < 1$, with probability at least $1 - \xi$,*

$$\sup_{h\in\mathcal{H}}|(Q_m - Q)h| \le 2\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{H}) + 6B\sqrt{\frac{\log(2/\xi)}{2m}}, \tag{B.11}$$

where the empirical complexity uses the absolute supremum as in (B.10)*.*

Proof. Put $Z = \sup_h |(Q_m - Q)h|$ and $R = \widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{H})$. Symmetrization gives $\mathbb{E}Z \le 2\mathbb{E}R$ [3, 33]. Changing one observation changes each of Z, R by at most $2B/m$. With $t = 2B\sqrt{\log(2/\xi)/(2m)}$, McDiarmid's inequality bounds each of $\mathbb{P}(Z > \mathbb{E}Z + t)$ and $\mathbb{P}(\mathbb{E}R > R + t)$ by $\xi/2$. Thus $Z \le 2R + 3t$ with probability at least $1 - \xi$, proving (B.11) for signed classes with $|h| \le B$. □

Conditional on D_{-k}, the observations in I_k are independent with law P, and Φ_k^σ is fixed with envelope B. Apply Lemma B.3 with $m = n_k$ and $\xi = \delta/(2K)$. Its conditional failure probability is at most $\delta/(2K)$, so the same bound holds unconditionally. A union bound over the $2K$ fold–endpoint pairs yields an event $\mathcal{E}_{\mathrm{emp}}$ of probability at least $1 - \delta$ on which

$$\begin{aligned}\sup_{f,\eta}\left|\sum_{k=1}^K w_k(P_{n,k} - P)\phi_{f,\eta}^{\sigma,(k)}\right| &\le 2\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^\sigma + 6B\sum_{k=1}^K w_k\sqrt{\frac{\log(4K/\delta)}{2n_k}} \\ &\le 2\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^\sigma + 6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}} \end{aligned} \tag{B.12}$$

for both endpoints. Here $P\phi_{f,\eta}^{\sigma,(k)}$ integrates the fitted function against a fresh observation. The training sets overlap; no independence between folds or between their success events is used. In particular, the proof never conditions simultaneously on all fitted nuisances, which could destroy the needed held-out independence.

B.3 Uniform endpoint estimation and complexity rates

Define the cross-fitted empirical objectives and their infima by

$$\begin{aligned}\widehat{Q}_f^\sigma(\eta) &= \eta + \frac{1}{\beta}\sum_{k=1}^K w_k P_{n,k}\phi_{f,\eta}^{\sigma,(k)}, \\ \widehat{\rho}^\sigma(f) &= \inf_{\eta\in[0,1]} \widehat{Q}_f^\sigma(\eta).\end{aligned} \tag{B.13}$$

Proof of Theorem 2. Under the theorem's assumptions and the measurability convention above, let $\mathcal{E}_\nu$ be the simultaneous nuisance event

$$\max_k \|\widehat{p}^{(-k)} - p\|_2 \le r_p, \qquad \max_k \|\widehat{\pi}^{(-k)} - \pi\|_2 \le r_\pi, \tag{B.14}$$

with probability at least $1 - \delta_\nu$. For $\delta \in (0,1)$ and $\delta_\nu \in [0, 1-\delta)$, the explicit calibrated radius is

$$\begin{aligned}r_n(\delta) = \frac{1}{\beta}\Bigg[2\max_{\sigma\in\{L,U\}}\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^\sigma + 6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}} \\ + \Gamma(\Gamma-1)r_p^2 + \frac{2\Gamma}{\pi_{\min}}r_p r_\pi\Bigg].\end{aligned} \tag{B.15}$$

On $\mathcal{E}_\nu$, Lemma B.2, applied to each realized pair of clipped fits, implies

$$\sup_{f,\eta,\sigma}\left|\sum_{k=1}^K w_k P\phi_{f,\eta}^{\sigma,(k)} - m_{f,\eta}^\sigma\right| \le \Gamma(\Gamma-1)r_p^2 + \frac{2\Gamma}{\pi_{\min}}r_p r_\pi.$$

The intersection $\mathcal{E} = \mathcal{E}_\nu \cap \mathcal{E}_{\mathrm{emp}}$ has probability at least $1 - \delta - \delta_\nu$ by the union bound. On this event, (B.12) and the last display give

$$\sup_{f,\eta,\sigma}\left|\sum_{k=1}^K w_k P_{n,k}\phi_{f,\eta}^{\sigma,(k)} - m_{f,\eta}^\sigma\right| \le \beta r_n(\delta).$$

Applying the infimum stability inequality (A.13) to $\widehat{Q}_f^\sigma$ and its population counterpart yields, on the same event of probability at least $1 - \delta - \delta_\nu$,

$$\sup_{f\in\mathcal{F}}\max_{\sigma\in\{L,U\}}|\widehat{\rho}^\sigma(f) - \rho^\sigma(f)| \le r_n(\delta). \tag{B.16}$$

This requires neither convexity of the signed empirical objective nor a unique threshold or loss density there. □

For a total confidence level $1-\delta_{\mathrm{tot}}$, allocate $\delta+\delta_\nu \le \delta_{\mathrm{tot}}$. The empirical complexity in (B.15) is an expectation over all Rademacher signs, conditional on the realized data. A Monte Carlo estimate is not automatically an upper bound for this expectation. Any random complexity or nuisance-error bound used instead must hold simultaneously on a specified validity event, whose failure probability is included in the confidence allocation.

We next bound the score complexity. For a sample $x_1,\dots,x_m$, let $\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{V})$ denote its absolute empirical complexity. Define

$$\mathcal{L}_y=\{x\mapsto \ell(f(x),y): f\in\mathcal{F}\},\qquad \mathcal{A}_y=\{x\mapsto(\ell(f(x),y)-\eta)_+ : f\in\mathcal{F},\ \eta\in[0,1]\}.$$

Define the threshold-augmented loss-class complexity by

$$\mathfrak{C}_n=\sum_{k=1}^{K} w_k\left\{\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_0)+\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_1)+n_k^{-1/2}\right\}, \tag{B.17}$$

where each term uses the covariates in its evaluation fold. For fixed fitted nuisances and $o=(x,s,y^{\mathrm{obs}})$, put $p_c=\widehat{p}^{(-k)}(x)$ and $w=(1-\widehat{\pi}^{(-k)}(x))/\widehat{\pi}^{(-k)}(x)$. The score is $\Psi_o^\sigma(a^0_{f,\eta}(x),a^1_{f,\eta}(x))$, where

$$\begin{aligned}\Psi_o^\sigma(u_0,u_1)=\ & s\{(1-y^{\mathrm{obs}})u_0+y^{\mathrm{obs}}u_1\}\\ &+(1-s)\{u_0+H^\sigma_{p_c}(u_1-u_0)\}\\ &+swG^\sigma_{p_c}(u_1-u_0)(y^{\mathrm{obs}}-p_c).\end{aligned} \tag{B.18}$$

It vanishes at $(0,0)$. Lemma B.1, $w\le 2/\pi_{\min}$, and $|\Delta u_1-\Delta u_0|\le 2\|\Delta u\|_\infty$ show that it is $L_*=3+4\Gamma/\pi_{\min}$-Lipschitz in the maximum norm, and hence also in Euclidean norm.

The vector-contraction inequality [24] therefore gives, for a universal constant C,

$$\begin{aligned}\widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}}\le CL_*\Big\{&\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_0)+\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_1)\\ &+n_k^{-1/2}\Big\}.\end{aligned} \tag{B.19}$$

To handle the absolute supremum, use that the score class contains zero at $\eta=1$: the absolute supremum is at most the sum of the positive and negative suprema, whose expectations agree by sign symmetry. Vector contraction on the positive supremum, followed by separating the two coordinate suprema, bounds it by a universal multiple of $L_*\sum_y\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{A}_y)$. Apply the same argument to the $\sqrt{2}$-Lipschitz hinge map $(z,t)\mapsto(z-t)_+$, adjoining zero to its input class if needed. Its coordinate complexities are those of $\mathcal{L}_y$ and the constant threshold class; the latter is $m^{-1}\mathbb{E}_\varepsilon|\sum_i\varepsilon_i|\le m^{-1/2}$. This proves (B.19). All contraction maps are continuous endpoint values or derivatives, including at ties; no discontinuous selector class is introduced.

If $\mathcal{L}_y$ has VC-subgraph dimension at most $d\ge 1$, then $\log N(\epsilon,\mathcal{L}_y,L_2(Q))\le C_0 d\log(C_1/\epsilon)$ for $0<\epsilon<1$ and every probability measure Q. After adjoining zero and the negative class, the standard entropy-to-complexity bound [3, 33] gives, uniformly over samples,

$$\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{L}_y)\le C'\sqrt{\frac{d+1}{m}}. \tag{B.20}$$

It therefore applies to the fixed fitted nuisances in each conditional argument.

For fixed K and balanced folds, $n_{\min}\ge c_K n$ for a positive constant c_K once all folds are nonempty. Equations (B.19)–(B.20) give

$$\begin{aligned}&\frac{1}{\beta}\left[2\max_\sigma\widehat{\mathfrak{R}}^\sigma_{n,\mathrm{abs}}+6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}}\right]\\ &\qquad\le\frac{C_{\Gamma,\pi_{\min},K}}{\beta}\sqrt{\frac{d+\log(1/\delta)}{n}}.\end{aligned} \tag{B.21}$$

For fixed $\Gamma,\pi_{\min},\beta$, nuisance rates $o(n^{-1/4})$ make the nuisance contribution in (B.15) $o(n^{-1/2})$. Joint high-probability nuisance bounds remain necessary; cross-fitting does not establish them. All uniformity is over f,η at fixed α,Γ.

Example B.4 (Conditional sparse-nuisance specialization)**.** Suppose sparse logistic nuisance fits in D covariates satisfy simultaneous $L_2(P_X)$ error bounds of orders

$$r_p \lesssim \sqrt{\frac{s_p \log(eD)}{n_{\mathrm{sel}}}}, \qquad r_\pi \lesssim \sqrt{\frac{s_\pi \log(eD)}{n_{\mathrm{tr}}}},$$

where s_p, s_π are sparsities, n_{tr} is the smallest training-fold size, and $n_{\mathrm{sel}} > 0$ is a simultaneous lower bound for the number of revealed training labels in every fold. Substitution in the proved remainder gives, up to the constants in these assumed rates,

$$\frac{1}{\beta}\left\{\Gamma(\Gamma-1)\frac{s_p \log(eD)}{n_{\mathrm{sel}}} + \frac{2\Gamma}{\pi_{\min}}\frac{\sqrt{s_p s_\pi}\log(eD)}{\sqrt{n_{\mathrm{sel}} n_{\mathrm{tr}}}}\right\}.$$

For a fixed number of balanced folds with nonempty training sets, $n_{\mathrm{tr}} \asymp n$. These substitutions require the stated model, design, tuning, and confidence conditions. A selected-population bound transfers to P_X because, for every measurable e,

$$\|e\|^2_{L_2(P_X)} \le \frac{P(S=1)}{\pi_{\min}}\|e\|^2_{L_2(P_{X\mid S=1})}. \tag{B.22}$$

Indeed, $\mathbb{E}[\pi(X)e(X)^2] = P(S=1)\mathbb{E}[e(X)^2 \mid S=1]$ and $\pi(X) \ge \pi_{\min}$. Learning p from revealed outcomes must therefore control target-population error.

Proposition B.5 (Finite threshold search for the fitted objectives)**.** *For a fixed f and realized nuisance fits, each objective in* (B.13) *attains its minimum on*

$$T_n(f) = \{0,1\} \cup \{\ell(f(X_i), y) : 1 \le i \le n,\ y \in \{0,1\}\}.$$

Proof. Thresholding preserves the label-loss ordering at each X_i. Thus $H^\sigma_{p_c}(d_{f,\eta})$ and $G^\sigma_{p_c}(d_{f,\eta})$ are fixed coefficients times $d_{f,\eta}$ throughout $[0,1]$; at $d_{f,\eta} = 0$ both representations vanish. Each score is a fixed linear combination of the two label-specific hinges, making the objective continuous and affine between successive knots of $T_n(f)$. Each such interval has an endpoint minimizer, even if the objective is constant there. Signed correction coefficients may prevent convexity but preserve this result. □

An exact implementation can sort the at most $2n+2$ breakpoints once per predictor. Write the fitted score as $c^\sigma_{i0}(\ell(f(X_i),0)-\eta)_+ + c^\sigma_{i1}(\ell(f(X_i),1)-\eta)_+$, with coefficients fixed across η. Starting from the objective at zero, sweep the sorted breakpoints: the objective is updated by its current slope times the distance to the next breakpoint, and the slope increases by $c^\sigma_{iy}/(n\beta)$ when the corresponding hinge vanishes. Combine all coefficients at a tied breakpoint and initialize the right slope using only losses above zero. Checking all knot values gives both global minima in $O(n \log n)$ time and $O(n)$ storage, after the fitted probabilities and label-specific losses have been evaluated.

B.4 Outer certificates, comparisons, and predictor selection

Proof of the decision guarantees in Section 4.2. Work on the joint event (B.16), of probability at least $1-\delta-\delta_\nu$, and use the certificate (8). Since $0 \le \rho^L(f) \le \rho^U(f) \le 1$, clipping preserves containment of the sharp interval and hence every compatible CVaR. Containment gives the lower width bound; clipping decreases width, while

$$\widehat{\rho}^U(f) - \widehat{\rho}^L(f) + 2r_n \le \rho^U(f) - \rho^L(f) + 4r_n,$$

proving the upper bound. If $\widehat{\rho}^U(f) + r_n < \widehat{\rho}^L(g) - r_n$, then for every compatible law,

$$\begin{aligned}\rho_{\alpha,P^\star}(f) &\le \rho^U(f) \le \widehat{\rho}^U(f) + r_n \\ &< \widehat{\rho}^L(g) - r_n \le \rho^L(g) \le \rho_{\alpha,P^\star}(g).\end{aligned}$$

Finally, any measurable $\varepsilon_{\mathrm{opt}}$-approximate minimizer $\widehat{f} \in \mathcal{F}$ of $\widehat{\rho}^U$ satisfies

$$\begin{aligned}\rho^U(\widehat{f}) &\le \inf_f \widehat{\rho}^U(f) + \varepsilon_{\mathrm{opt}} + r_n \\ &\le \inf_f \rho^U(f) + \varepsilon_{\mathrm{opt}} + 2r_n.\end{aligned}$$

The last inequality takes infima of the pointwise endpoint bound, so no exact minimizer is needed. Uniformity permits all predictors in these claims to be chosen using the same data. □

The selection benchmark is the smallest compatible upper endpoint in the class. For every compatible $P^\star$, the same argument gives $\rho_{\alpha,P^\star}(\widehat{f}) \le \inf_f \rho^U(f) + 2r_n + \varepsilon_{\mathrm{opt}}$. Clipped endpoint errors are at most $2r_n$ on the same event. Convergence in probability follows if $r_n \to 0$ in probability and $\delta_n + \delta_{\nu,n} \to 0$; a fixed failure probability remains fixed.

B.5 Oracle calibration for finite-cell experiments

For the oracle coverage experiment, suppose $p = p_1$ is known and define

$$\begin{aligned} h^\sigma_{f,\eta}(O) &= S a^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1-S) b^\sigma_{f,\eta}(X; p(X)), \\ \widehat{\rho}^\sigma_{\mathrm{or}}(f) &= \inf_{\eta\in[0,1]} \{\eta + \beta^{-1} P_n h^\sigma_{f,\eta}\}. \end{aligned} \tag{B.23}$$

This score lies in $[0,1]$ and has mean $m^\sigma_{f,\eta}$; estimating π is unnecessary for this oracle construction.

Proposition B.6 (Finite-cell oracle certificate)**.** *Fix a predictor f independently of the evaluation observations. Suppose P_X is supported on J covariate values $x_1, \dots, x_J$. Let*

$$T(f) = \{0,1\} \cup \{\ell(f(x_j), y) : 1 \le j \le J,\ y \in \{0,1\}\}, \qquad M = |T(f)| \le 2J + 2.$$

With probability at least $1-\delta$,

$$\max_{\sigma\in\{L,U\}} |\widehat{\rho}^\sigma_{\mathrm{or}}(f) - \rho^\sigma(f)| \le r_{\mathrm{or}}(\delta) := \frac{1}{\beta}\sqrt{\frac{\log(4M/\delta)}{2n}}. \tag{B.24}$$

Thus the clipped interval obtained from (8) using these oracle estimators and r_{or} contains the entire sharp population interval with probability at least $1-\delta$.

Proof. By the piecewise-affine argument of Proposition B.5, both $h^\sigma_{f,\eta}(O)$ and its expectation have knots only in $T(f)$. The absolute empirical-minus-population difference is maximal at a knot:

$$\sup_{\eta\in[0,1]} |(P_n - P) h^\sigma_{f,\eta}| = \max_{\eta\in T(f)} |(P_n - P) h^\sigma_{f,\eta}|.$$

Hoeffding's inequality and a union bound over the $2M$ endpoint–threshold pairs bound the failure probability at tolerance u by $4Me^{-2nu^2}$. Taking $u = \sqrt{\log(4M/\delta)/(2n)}$ and applying (A.13) proves (B.24); The containment argument in Appendix B.4 gives whole-interval coverage. □

The population-support set $T(f)$ includes losses at cells absent from a particular sample; observed knots $T_n(f)$ suffice for empirical minimization but cannot replace $T(f)$ in this concentration argument. Conditioning covers a predictor trained on independent data. Selection using evaluation data requires a uniform bound: for a fixed finite family $\mathcal{F}_0$, replace M by $\sum_{f\in\mathcal{F}_0} |T(f)|$. Known p supplies the oracle experiment's premise; coverage for learned-nuisance estimators instead requires Theorem 2's controls.

B.6 Revealed-tail learning lower bound

Equal tail-cell masses and constant observation probability specify the lower-bound construction; the general model allows covariate-dependent observation and heterogeneous revelation costs.

Theorem B.7 (Revealed-tail observation complexity)**.** *Fix $\alpha \in (0,1)$, $\beta = 1-\alpha$, $\pi_0 \in (0,1]$, and $n \ge 1$. For every integer $1 \le d \le n\pi_0\beta$, there exist a binary classification class $\mathcal{F}$ of VC dimension d and a family $\mathcal{P}$ of full-data laws with $\Gamma = 1$ and $\pi(X) = \pi_0$ such that*

$$\begin{aligned} &\inf_{\widehat{f}} \sup_{P^\star\in\mathcal{P}} \mathbb{E}_{(P^\star_{\mathrm{obs}})^{\otimes n}} \left[\rho_{\alpha,P^\star}(\widehat{f}) - \inf_{f\in\mathcal{F}} \rho_{\alpha,P^\star}(f)\right] \\ &\qquad \ge \frac{1}{16}\sqrt{\frac{d}{n\pi_0\beta}}. \end{aligned} \tag{B.25}$$

The infimum ranges over measurable learners taking values in $\mathcal{F}$, including randomized learners.

Proof. We use a direct binary-testing form of the hypercube argument [34]. Fix $\pi_0 \in (0,1]$, $n \geq 1$, and an integer $1 \leq d \leq n\pi_0\beta$. Put

$$\gamma = \frac{1}{8}\sqrt{\frac{d}{n\pi_0\beta}} \leq \frac{1}{8}. \tag{B.26}$$

Let $\mathcal{X} = \{\star, 1, \ldots, d\}$ with $P_X(\star) = 1-\beta$ and $P_X(j) = \beta/d$. For $v \in \{-1,1\}^d$, define $f_v(\star) = 0$ and $f_v(j) = (1+v_j)/2$, and set $\mathcal{F} = \{f_v : v \in \{-1,1\}^d\}$. The class shatters $\{1,\ldots,d\}$, and cannot shatter any larger set because its value at $\star$ is fixed. Its VC dimension is exactly d. Use ordinary zero-one loss $\ell(f(x), y) = \mathbf{1}\{f(x) \neq y\}$.

For each $\theta \in \{-1,1\}^d$, construct a full-data law P_θ with $Y = 0$ at $\star$ and

$$P_\theta(Y = 1 \mid X = j) = \frac{1}{2} + \gamma\theta_j, \qquad j = 1, \ldots, d.$$

Let $S \sim \text{Bernoulli}(\pi_0)$ independently of (X, Y), and observe (X, S, SY). These laws satisfy overlap with $\pi(X) = \pi_0$ and conditional ignorability, hence belong to the $\Gamma = 1$ model. All losses vanish at $\star$. The conditional error within $\{1,\ldots,d\}$ is

$$R_\theta(f_v) = \frac{1}{2} - \gamma + \frac{2\gamma}{d}\sum_{j=1}^{d} \mathbf{1}\{v_j \neq \theta_j\}. \tag{B.27}$$

The full-population loss is Bernoulli with parameter $q_{\theta,v} = \beta R_\theta(f_v) \leq \beta$; the Bernoulli identity (A.5) gives

$$\begin{aligned} \rho_{\alpha,P_\theta}(f_v) &= R_\theta(f_v), \\ \rho_{\alpha,P_\theta}(f_v) - \inf_{f\in\mathcal{F}} \rho_{\alpha,P_\theta}(f) &= \frac{2\gamma}{d}\sum_{j=1}^{d} \mathbf{1}\{v_j \neq \theta_j\}. \end{aligned} \tag{B.28}$$

The minimizing predictor is f_θ.

Let $Q_\theta = P_{\theta,\text{obs}}^{\otimes n}$ and let $\theta^{(j)}$ be obtained by reversing the jth coordinate. For one observation, the two distributions differ only on $\{X = j, S = 1\}$, which has probability $\beta\pi_0/d$ under both laws. Conditioning on (X, S) and using the chain rule for relative entropy therefore gives

$$\begin{aligned} D_{\text{KL}}(Q_\theta \| Q_{\theta^{(j)}}) &= \frac{n\beta\pi_0}{d} D_{\text{KL}}(\text{Bernoulli}(1/2+\gamma) \,\|\, \text{Bernoulli}(1/2-\gamma)) \\ &\leq \frac{16 n\beta\pi_0\gamma^2}{d} = \frac{1}{4}. \end{aligned} \tag{B.29}$$

Here the symmetric Bernoulli divergences both equal $2\gamma\log\{(1+2\gamma)/(1-2\gamma)\} \leq 16\gamma^2$ for $\gamma \leq 1/4$, and independence supplies the factor n. Pinsker's inequality gives

$$\text{TV}(Q_\theta, Q_{\theta^{(j)}}) \leq \sqrt{D_{\text{KL}}(Q_\theta \| Q_{\theta^{(j)}})/2} \leq \frac{1}{2\sqrt{2}} \leq \frac{1}{2}. \tag{B.30}$$

Consider any measurable learner taking values in $\mathcal{F}$, written $\widehat{f} = f_{\widehat{v}}$. For a fixed coordinate j, pair each θ with $\theta_j = 1$ with $\theta^{(j)}$, and write $A_j = \{\widehat{v}_j = 1\}$. The sum of the two testing errors satisfies

$$\begin{aligned} Q_\theta(\widehat{v}_j \neq \theta_j) + Q_{\theta^{(j)}}(\widehat{v}_j \neq \theta_j^{(j)}) &= Q_\theta(A_j^c) + Q_{\theta^{(j)}}(A_j) \\ &\geq 1 - \text{TV}(Q_\theta, Q_{\theta^{(j)}}) \geq \frac{1}{2}. \end{aligned}$$

There are 2^{d-1} such pairs. Averaging over all 2^d vertices gives $2^{-d}\sum_\theta Q_\theta(\widehat{v}_j \neq \theta_j) \geq 1/4$. Sum this inequality over j and use (B.28) to obtain

$$\frac{1}{2^d}\sum_\theta \mathbb{E}_{Q_\theta}\left[\rho_{\alpha,P_\theta}(\widehat{f}) - \inf_{f\in\mathcal{F}} \rho_{\alpha,P_\theta}(f)\right] \geq \frac{2\gamma}{d}\frac{d}{4} = \frac{1}{16}\sqrt{\frac{d}{n\pi_0\beta}}.$$

The supremum over θ is at least this average, for every learner. Taking the infimum over learners proves Theorem B.7. Randomized learners satisfy the same bound: append their independent random seed to each experiment, which leaves the neighboring total variation unchanged, and apply the identical testing argument. □

Only the expected $n\pi_0\beta$ revealed observations in the tail stratum distinguish the vertices. This is an expected excess-CVaR lower bound for $\mathcal{F}$-valued learners, already with fully identified risks at $\Gamma = 1$.

C Revelation Theory and Algorithms

Fix f, $\alpha \in (0,1)$, $\beta = 1-\alpha$, and finite $\Gamma \geq 1$. Use the historical π, p_1, sensitivity bounds $[\underline{p}_\Gamma, \overline{p}_\Gamma]$, and $s_\Gamma = \overline{p}_\Gamma - \underline{p}_\Gamma$. Write $\mu = P_X$ and $\mathbb{E} = \mathbb{E}_\mu$. All kernels and policies are measurable; conditional statements apply on components of positive population mass.

C.1 Retained-information model and compatible laws

Additional revelation $A \in \{0,1\}$ satisfies $A \leq 1-S$, with intensity $\lambda(x) = P(A=1 \mid X = x, S=0) \in [0,1]$. The audited population law has known $p_A(x) = P(Y=1 \mid X=x, A=1)$ satisfying

$$\underline{p}_\Gamma \leq p_A \leq \overline{p}_\Gamma \quad \text{on } \{(1-\pi)\lambda > 0\}. \tag{C.1}$$

Extend p_A within these bounds on its zero-mass region; this has no effect on risks.

Definition C.1 (Retained-sensitivity compatibility)**.** The class $\mathcal{P}^{\mathrm{ret}}_{\Gamma,\lambda}$ consists of laws of (X,S,A,Y) with X-marginal μ and conditional components

(S,A)	$P(S,A \mid X=x)$	$P(Y=1 \mid X=x,S,A)$
$(1,0)$	$\pi(x)$	$p_1(x)$
$(0,1)$	$(1-\pi(x))\lambda(x)$	$p_A(x)$
$(0,0)$	$(1-\pi(x))(1-\lambda(x))$	$q(x)$.

The residual kernel ranges freely over $q \in [\underline{p}_\Gamma, \overline{p}_\Gamma]$; $(S,A) = (1,1)$ has zero mass.

Lemma C.2 (Historical compatibility)**.** *Every admissible q defines a retained law whose marginal belongs to $\mathcal{P}_\Gamma(P_{\mathrm{obs}})$, with original hidden kernel*

$$p_0 = \lambda p_A + (1-\lambda)q \quad \textit{on } \{\pi < 1\}. \tag{C.2}$$

The retained class is nonempty.

Proof. Apply the kernel construction of Appendix A.1 to the table. Its masses sum to one and its selected component is unchanged. Conditional total probability gives (C.2); convexity of the sensitivity interval proves compatibility. Taking $q = \underline{p}_\Gamma$ proves nonemptiness. □

At $\lambda = 0$ the historical class is recovered; at $\lambda = 1$ no residual kernel remains. Membership requires the audited and residual bounds separately: a bound on their mixture p_0 alone does not imply either component bound.

Known randomization $A \perp Y \mid (X, S=0)$ supplies extra information: $p_A = p_0$ on the audited support, and $q = p_0$ wherever residual mass remains. Thus the population audited law identifies p_0 wherever originally hidden cases have positive audit probability. The retained class instead keeps its residual kernel free within the displayed bounds. Finite audits additionally require estimation and uncertainty control for the audited contribution; a positive population audit probability alone supplies no finite-sample precision guarantee.

C.2 Post-revelation endpoints and tail identification value

Use $a^y_{f,\eta}, d_{f,\eta}, b^\sigma_{f,\eta}$ from Section 3, write $s_\Gamma = \overline{p}_\Gamma - \underline{p}_\Gamma$, and set $w_\lambda = (1-\pi)(1-\lambda)$. The known contribution and a residual kernel's threshold objective are

$$K_{f,\eta}(\lambda) = \mathbb{E}[\pi(a^0_{f,\eta} + p_1 d_{f,\eta}) + (1-\pi)\lambda(a^0_{f,\eta} + p_A d_{f,\eta})], \tag{C.3}$$

$$G_q(\eta) = \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda(a^0_{f,\eta} + q d_{f,\eta})]\}. \tag{C.4}$$

Conditioning on the table and optimizing the affine residual term give

$$\begin{aligned} L_{f,\eta}(\lambda) &= \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda b^L_{f,\eta}]\}, \\ U_{f,\eta}(\lambda) &= \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda b^U_{f,\eta}]\}. \end{aligned} \tag{C.5}$$

Proposition C.3 (Sharp retained-sensitivity interval)**.** *The attainable population risks are exactly*

$$\{\rho_{\alpha,Q}(f) : Q \in \mathcal{P}^{\mathrm{ret}}_{\Gamma,\lambda}\} = [\min_{\eta\in[0,1]} L_{f,\eta}(\lambda), \min_{\eta\in[0,1]} U_{f,\eta}(\lambda)].$$

Both threshold minima and endpoint laws are attained.

Proof. On $H = \{x : \pi(x) < 1\}$, the common kernels q^L_f, q^U_f of Appendix A.3 satisfy $a^0_{f,\eta} + q^\sigma_f d_{f,\eta} = b^\sigma_{f,\eta}$ for every $\eta \in [0,1]$. On H^c, the residual weight $w_\lambda = (1-\pi)(1-\lambda)$ is zero. Thus $L_{f,\eta}(\lambda) \le G_q(\eta) \le U_{f,\eta}(\lambda)$ for every admissible q, with equality at the corresponding common kernel. Taking minima proves the endpoint claims; continuity of each hinge objective on $[0,1]$ gives attainment. For $q_t = (1-t)q^L_f + tq^U_f$,

$$\sup_\eta |G_{q_t}(\eta) - G_{q_s}(\eta)| \le |t-s|\beta^{-1}\mathbb{E}[w_\lambda s_\Gamma].$$

The infimum inequality (A.13) makes the optimized risk continuous in t. This admissible path joins the endpoints and fills their interval. □

Proposition C.4 (Exact revelation value)**.** *With*

$$\tau_{f,\eta} = (1-\pi)s_\Gamma|d_{f,\eta}|/\beta, \tag{C.6}$$

$$W_{f,\eta}(\lambda) := U_{f,\eta}(\lambda) - L_{f,\eta}(\lambda) = \mathbb{E}[(1-\lambda)\tau_{f,\eta}] = W_{f,\eta}(0) - \mathbb{E}[\lambda\tau_{f,\eta}], \tag{C.7}$$

every bounded measurable h and feasible t satisfy

$$W_{f,\eta}(\lambda + th) - W_{f,\eta}(\lambda) = -t\mathbb{E}[h\tau_{f,\eta}]. \tag{C.8}$$

Proof. Subtract (C.5): K cancels and $b^U - b^L = s_\Gamma|d|$. Substitution gives the perturbation identity, including one-sided feasible directions. Since $0 \le \tau \le \beta^{-1}$, all terms are integrable. The fixed-threshold reduction is independent of p_A, even when that law varies with the policy. □

C.3 Minimum-cost precision and the exact frontier

Fix f, η, Γ, abbreviate $\tau = \tau_{f,\eta}$, and let $c \ge 0$ be measurable with $\mathbb{E}[c] < \infty$. Set $T = \mathbb{E}[\tau] = W_{f,\eta}(0)$, $a = (T-\varepsilon)_+$ for $\varepsilon \ge 0$, and $F = \mathbb{E}[\tau \mathbf{1}_{\{c=0\}}]$. By (C.7), the precision constraint is $\mathbb{E}[\tau\lambda] \ge a$.

Proof of Theorem 3. We establish attainment and the primal–dual identity

$$\mathrm{Cert}_{f,\eta}(\varepsilon) = \min_{0\le\lambda\le1}\{\mathbb{E}[c\lambda] : \mathbb{E}[\tau\lambda] \ge a\} = \sup_{u\ge0}\{ua - \mathbb{E}[(u\tau - c)_+]\}. \tag{C.9}$$

Pointwise, $(c - u\tau)\lambda \ge -(u\tau - c)_+$, so every feasible policy obeys

$$\mathbb{E}[c\lambda] \ge u\mathbb{E}[\tau\lambda] - \mathbb{E}[(u\tau - c)_+] \ge ua - \mathbb{E}[(u\tau - c)_+]. \tag{C.10}$$

The following four cases attain this lower bound, or its limit.

No reduction ($a = 0$). Take $\lambda = 0, u = 0$, giving zero cost; this includes $T = 0$.

Free reduction ($0 < a \le F$). Take $\lambda = (a/F)\mathbf{1}_{\{c=0,\tau>0\}}$ and $u = 0$. The reduction is exactly a at zero cost, including when the free component has atoms.

Interior reduction ($F < a < T$). On $D = \{c > 0, \tau > 0\}$, use the finite measure $\nu(B) = \int_B \tau\, d\mu$ and $r = c/\tau$. Put $b = a - F \in (0, \nu(D))$ and choose

$$u_* = \inf\{v \ge 0 : \nu(D \cap \{r \le v\}) \ge b\}. \tag{C.11}$$

Because $r \in (0,\infty)$ ν-almost everywhere, $0 < u_* < \infty$. Continuity of a finite measure along monotone sets gives $H := \nu(D \cap \{r < u_*\}) \le b \le H + J$, where $J := \nu(D \cap \{r = u_*\})$. Set $\theta = (b - H)/J$ if $J > 0$ and $\theta = 0$ otherwise (then $H = b$). The policy

$$\lambda_* = \mathbf{1}_{\{c=0,\tau>0\}} + \mathbf{1}_{D\cap\{r<u_*\}} + \theta\mathbf{1}_{D\cap\{r=u_*\}} \tag{C.12}$$

achieves reduction $F + H + \theta J = a$ and equality in the pointwise dual bound. Its cost equals $u_*a - \mathbb{E}[(u_*\tau - c)_+]$. It ranks by τ/c with threshold $1/u_*$; fractional ties handle atoms without an atomlessness assumption.

Full reduction ($a = T > F$). Feasibility forces $\mathbb{E}[\tau(1-\lambda)] = 0$, hence $\lambda = 1$ almost surely on $\{\tau > 0\}$. The minimum cost is $\mathbb{E}[c\mathbf{1}_{\{\tau>0\}}]$, attained by revealing exactly that set, while

$$uT - \mathbb{E}[(u\tau - c)_+] = \mathbb{E}[\min\{u\tau, c\}] \uparrow \mathbb{E}[c\mathbf{1}_{\{\tau>0\}}]$$

by monotone convergence. Thus the dual supremum agrees even without a finite maximizer. Its shortfall is $\mathbb{E}[(c - u\tau)_+\mathbf{1}_{\{\tau>0\}}]$, so a finite optimizer exists exactly when $c \le u\tau$ almost surely on $\{\tau > 0\}$ for some finite u. Full and free reduction correspond to value-density thresholds 0 and ∞. Zero-value locations can always be left unselected; if $F = T$ the whole frontier is free. These cases exhaust $0 \le a \le T$. □

C.4 From threshold precision to optimized CVaR precision

Denote the endpoints in Proposition C.3 by $\underline{\rho}^{\mathrm{ret}}_{\alpha,\Gamma,\lambda}(f)$ and $\overline{\rho}^{\mathrm{ret}}_{\alpha,\Gamma,\lambda}(f)$, and their difference by $I^{\mathrm{ret}}_{\Gamma,\lambda}(f)$.

Proposition C.5 (Controlling the optimized CVaR interval)**.** *For the fixed predictor,*

$$0 \le I^{\mathrm{ret}}_{\Gamma,\lambda}(f) \le \sup_{\eta\in[0,1]} W_{f,\eta}(\lambda) = W_{f,0}(\lambda). \tag{C.13}$$

Theorem 3 at $\eta = 0$ therefore minimizes cost for the sufficient condition $W_{f,0}(\lambda) \le \varepsilon$.

Proof. If η_L minimizes L, then $\min_\eta U - \min_\eta L \le U_{f,\eta_L} - L_{f,\eta_L} = W_{f,\eta_L}$; nonnegativity follows from ordered endpoints. For $v = \max_y \ell(f(x), y)$ and $w = \min_y \ell(f(x), y)$,

$$|d_{f,\eta}(x)| = (v-\eta)_+ - (w-\eta)_+ = (v - \max\{w, \eta\})_+.$$

This decreases in η and equals $v - w$ at zero. Integration against $(1-\lambda)(1-\pi)s_\Gamma/\beta \ge 0$ proves the supremum identity. □

Direct optimized-width design. Fix an admissible set Λ_{adm} of measurable revelation intensities and a specified family of population audit laws $\{P^\lambda : \lambda \in \Lambda_{\mathrm{adm}}\}$ sharing one (X, S, Y)-marginal. Each P^λ has intensity λ and satisfies Definition C.1. Set $p^\lambda_A(x) = P^\lambda(Y = 1 \mid X = x, A = 1)$, with an extension within the sensitivity bounds on zero-mass regions. Write $I^{\mathrm{ret}}_{\Gamma,\lambda}(f; p^\lambda_A)$ for the width obtained using this kernel in (C.3), retaining the original free residual bounds. For the cost density c above and $\varepsilon \ge 0$, define

$$\mathrm{Cost}^{\mathrm{CVaR}}_f(\varepsilon) := \inf_{\lambda\in\Lambda_{\mathrm{adm}}} \left\{\mathbb{E}[c\lambda] : I^{\mathrm{ret}}_{\Gamma,\lambda}(f; p^\lambda_A) \le \varepsilon\right\}, \tag{C.14}$$

with $\inf \varnothing = +\infty$. The audited contribution can affect the separate threshold minima even though it cancels from $W_{f,\eta}(\lambda)$.

Strictness of the sufficient bound. Take constant X, zero-one loss, $f = 0$, $\pi = p_1 = p_A = 1/2$, $\lambda = 1/2$, $\Gamma = 2$, and $\beta = 1/4$. The original bounds are $[1/3, 2/3]$. Under the retained lower and upper endpoint laws, the probabilities r_L and r_U of unit loss $L_f = 1$ are

$$r_L = \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{12} = \tfrac{11}{24}, \qquad r_U = \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{6} = \tfrac{13}{24}.$$

Both exceed β, so (A.5) gives optimized endpoints one and $I^{\mathrm{ret}}_{\Gamma,\lambda}(f) = 0$, although $W_{f,0}(\lambda) = \{(1/2)(1/2)/(1/4)\}(2/3 - 1/3) = 1/3$. Thus the sufficient criterion can require revelation when optimized risk is already identified. Predictor reoptimization and estimated acquisition values require their own statistical or sequential analysis.

C.5 Finite-cell computation and operational cost units

For cells $j = 1, \ldots, K$ with masses $\mu_j > 0$, write $\tau_j = \tau_{f,\eta}(j)$, $c_j = c(j)$, and set $v_j = \mu_j\tau_j$, $w_j = \mu_j c_j$, $T = \sum_j v_j$, $a = (T - \varepsilon)_+$, and $F = \sum_{j:c_j=0} v_j$. The exact fractional algorithm is:

1. If $a = 0$, reveal nothing. If $0 < a \le F$, reveal fraction a/F of the zero-cost, positive-value component; both cases cost zero.
2. Otherwise reveal all useful free cells. Order the positive-cost, positive-value cells by decreasing τ_j/c_j; set $R_k = \sum_{j\le k} v_j$, $C_k = \sum_{j\le k} w_j$, and $R_0 = C_0 = 0$.

3. For $b = a - F$, let $k = \min\{j : R_j \geq b\}$. Reveal the cells before k, fraction $\theta = (b - R_{k-1})/v_k$ of cell k, and none after it. Then

$$\mathrm{Cert}_{f,\eta}(\varepsilon) = C_{k-1} + \theta w_k = C_{k-1} + \frac{b - R_{k-1}}{\tau_k / c_k}. \tag{C.15}$$

Tied ratios may be ordered arbitrarily; full reduction gives $\theta = 1$ in the last useful cell. If there are no positive-cost useful cells, the first step covers every feasible target. This is Theorem 3's policy, so the frontier is piecewise linear. For fixed cell policies and audited laws, the piecewise-affine argument of Proposition B.5 optimizes both retained objectives over $\{0,1\} \cup \{\ell(f(j), y) : 1 \leq j \leq K,\ y \in \{0,1\}\}$.

If a hidden label costs $\kappa(x) \geq 0$, the target-population cost density and operational value density are

$$c(x) = (1 - \pi(x))\kappa(x), \qquad \mathbb{E}[c\lambda] = \mathbb{E}[(1-\pi)\lambda\kappa], \tag{C.16}$$

$$\frac{\tau_{f,\eta}(x)}{c(x)} = \frac{s_\Gamma(x)|d_{f,\eta}(x)|}{\beta\kappa(x)} \quad \text{when } (1-\pi)\kappa > 0. \tag{C.17}$$

The hidden-mass factor cancels; unit label cost ranks already-hidden cases by $s_\Gamma|d|/\beta$. Cell masses likewise cancel from v_j/w_j. The algorithm allocates divisible population mass. Whole-individual batches add integer constraints and sampling variation. Sequential retraining changes losses and acquisition values; confidence guarantees across adaptive rounds additionally require control of newly revealed contributions, acquisition history, and stopping.

D Experimental Protocols and Results

This appendix specifies the experiments in Section 6: finite-cell identification and estimation, historical applicant assessment, controlled masking, and targeted revelation. All results use saved outputs; Appendix D.7 identifies the accompanying provenance.

D.1 Evaluation targets, interval construction, and aggregation

Write $\beta = 1 - \alpha$, $\rho^L = \underline{\rho}_{\alpha,\Gamma}$, $\rho^U = \overline{\rho}_{\alpha,\Gamma}$, and $I_\Gamma = \rho^U - \rho^L$. Endpoints minimize their respective threshold objectives. Finite-support minima are found exactly at 0, 1, and all label-specific loss breakpoints, including when estimated hinge coefficients are signed. Sampling and lending studies use Brier loss $(f - y)^2$; population sweeps and the synthetic frontier use absolute loss.

Study	Target	Metric	Replications/ aggregation
Finite-cell sampling	Sharp population interval; its endpoints	Oracle whole-interval containment; learned-endpoint RMSE	60 worlds × 80 samples per n; means/RMSE
Historical applicants	Applicant-level regression-extension sensitivity interval	Estimated endpoints and strict comparisons	One cohort; five-fold cross-fitting
Controlled masking	Full-outcome CVaR of each fitted predictor	Reference coverage; comparison errors and unresolved decisions	20 masks; counts and mean widths
Fixed revelation	Fixed-threshold retained gap $W_{f,\eta}$	Remaining gap; cost to target	120 synthetic worlds or 20 masks; within-run normalization, then medians/IQR
Sequential audits	Estimated optimized retained width; held-out predictive loss	Width, mean Brier loss, CVaR; paired TIV–random differences	20 masks; medians and descriptive IQR

For fixed f, η, the revelation target is $W_{f,\eta}(\lambda) = \mathbb{E}[(1-\lambda)\tau_{f,\eta}]$, where $\tau_{f,\eta} = (1-\pi)s_\Gamma|d_{f,\eta}|/\beta$. It differs from the optimized width, whose endpoints are minimized separately (Appendix C.4). Oracle containment, masking-reference coverage, and sequential plug-in precision have different

calibration premises. Fixed-audit reductions normalize within each run before aggregation; sequential widths aggregate the recorded widths. Ratios of separately aggregated values need not equal aggregated relative reductions.

D.2 Finite-cell identification and learned-endpoint accuracy

Generator and estimators. Each of 60 simulated populations has covariate support $\mathcal{X}_0$ consisting of 30 cells, with Dirichlet$(1, \ldots, 1)$ masses, independent $\pi \sim U[.2, .95]$ and $p_1 \sim U[.05, .95]$, and $p_0 \mid p_1 \sim U[q_-(p_1), q_+(p_1)]$ at $\Gamma = 2$.

Fix $f = \text{clip}_{[0,1]}(p_1 + N(0, .15^2))$ within each world, use Brier loss and $\alpha = .75$, and compute exact endpoints and generating-law CVaR on $T_f = \{0, 1\} \cup \{f(x)^2, (1 - f(x))^2 : x \in \mathcal{X}_0\}$. For each $n \in \{1000, 3000, 10000, 30000, 100000\}$, draw 80 samples per world (4,800 replications). The oracle uses known p_1 and the bounded score of Appendix B.5; its endpoints are expanded by

$$r_{\text{or}} = \frac{1}{\beta}\sqrt{\frac{\log(4|T_f|/\delta)}{2n}}, \qquad \delta = .05,$$

then clipped to $[0, 1]$. Proposition B.6 gives whole-interval containment with probability at least .95 for this fixed predictor. Saved generating-law CVaR coverage also equals one at every reported sample size.

On the same records, randomly assign each observation to one of two folds. In a given training cell, let N_{11} count records with $(S, Y) = (1, 1)$, let N_{10} count records with $(S, Y) = (1, 0)$, and let N be the total number of training records in that cell. Estimate

$$\widehat{p}_1 = \frac{N_{11} + 1/2}{N_{11} + N_{10} + 1}, \qquad \widehat{\pi} = \frac{N_{11} + N_{10}}{N},$$

clipped to $[.01, .99]$ and $[.1, 1]$, respectively; an empty training cell receives propensity $1/2$. Evaluate the locally robust score using the other fold's fits. The resulting RMSEs measure endpoint accuracy. Learned-nuisance coverage requires a separate calibration argument.

n	Sharp width	Radius	Outer width	Coverage	RMSE upper	RMSE lower
1,000	.0949	.260	.614	1.000	.0361	.0425
3,000	.0949	.150	.395	1.000	.0171	.0160
10,000	.0949	.082	.259	1.000	.0087	.0076
30,000	.0949	.047	.190	1.000	.0049	.0043
100,000	.0949	.026	.147	1.000	.0027	.0023

Table 1: Repeated-sampling decomposition over 4,800 replications per sample size. Widths and oracle radii are means; coverage is the proportion of oracle outer intervals containing both exact endpoints. RMSE uses two-fold learned-nuisance estimates on those same sampled records.

Observed-tail scaling. For the $d = 16$ tail-cell construction of Appendix B.6, use 240 replications at each of 54 settings: $n \in \{600, 900, 1400, 2200, 3500, 5500\}$, $\beta \in \{.10, .15, .25\}$, and $\pi_0 \in \{.20, .35, .55\}$. Set $\gamma = \min\{.20, .16\sqrt{d/(n\beta\pi_0)}\}$ and recover each cell's sign by majority vote among revealed tail labels, assigning ties the positive sign. The metric is 2γ times the incorrectly recovered fraction, equal to excess CVaR for this construction. Its least-squares guide against $1/\sqrt{n\pi_0\beta}$ has slope .5030, intercept $-.0018$, and $R^2 \simeq .998$; this illustration does not establish a general matching upper bound for Appendix B.6.

Population sweeps. The five-region absolute-loss design has region masses $\mu_j = P(X = j)$ and full-data outcome probabilities $p_{\text{full},j} = P(Y = 1 \mid X = j)$, for $j = 1, \ldots, 5$. It fixes

$$\mu = (.45, .15, .15, .10, .15), \quad p_{\text{full}} = (.04, .75, .10, .40, .50),$$
$$s_1 = (.80, .03, .80, .80, .80), \quad s_0 = (.80, .60, .80, .80, .80),$$
$$f = (.05, .25, .10, .40, .50), \quad s_y = P(S = 1 \mid X, Y = y).$$

Exact finite-support optimization evaluates 60 tail levels on $[.55, .95]$ at $\Gamma = 50$, and 60 common historical observation probabilities on $[.05, 1]$ at $\alpha = .85, \Gamma = 50$, holding the original p_1 fixed in the latter sweep. An archived sweep uses 60 log-spaced $\Gamma \in [1, 100]$ at $\alpha = .85$; values below the generating sensitivity shift do not claim generating-law coverage.

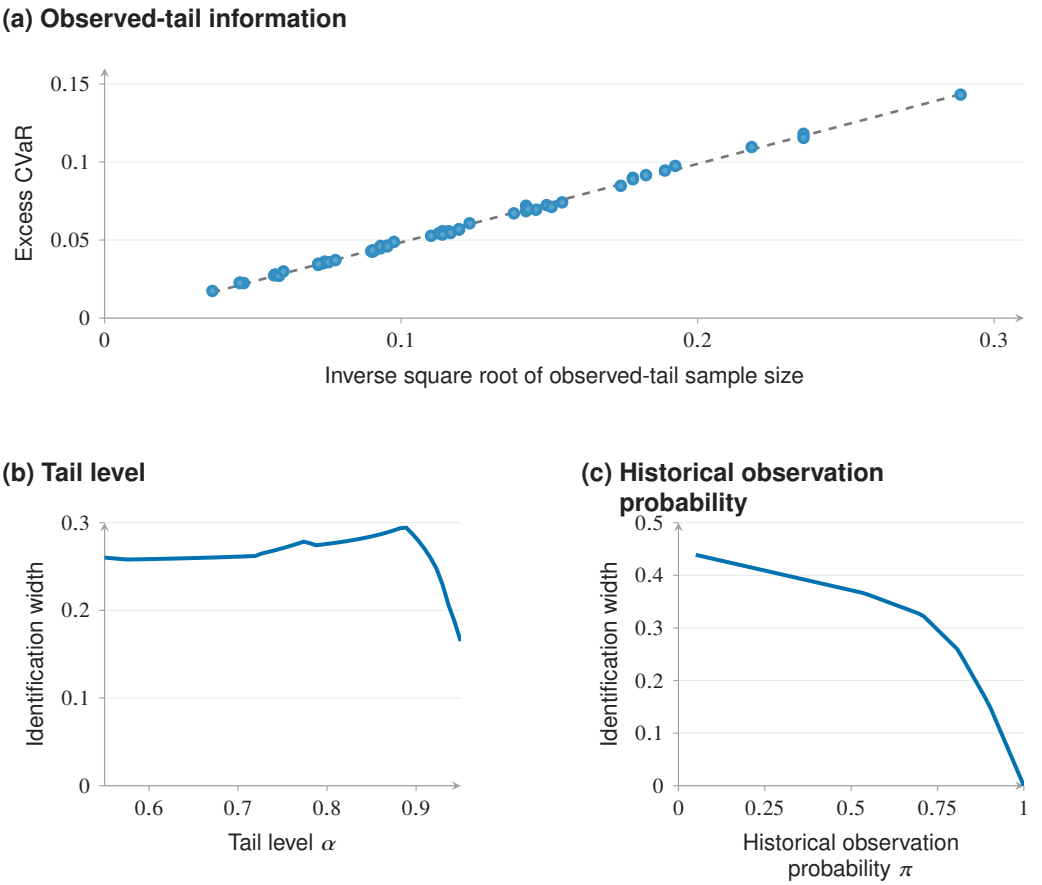


Figure 5: **Observation scarcity and sensitivity shape tail-risk precision.** (a) Recorded simulations of the Appendix B.6 learning construction, with a least-squares guide against $(n\pi_0\beta)^{-1/2}$. Here $n\pi_0\beta$ is the expected observed-tail sample count. (b–c) Exact identification widths for the fixed five-region design as the tail level or historical observation probability varies. Width can vary nonmonotonically with α. Lines join recorded evaluations.

D.3 Historical LendingClub applicant assessment

Cohort and shared preprocessing. From the public 2007–2018 accepted/rejected files [13], retain loans issued and applications rejected in 2012. Mature issued outcomes are charged-off/default ($Y = 1$) or fully paid ($Y = 0$), including corresponding former-credit-policy statuses; exclude nonmature outcomes. The saved cohort has 390,644 applications: 53,367 issued loans (8,644 defaults) and 337,277 rejections. Repayment under issuance is unobserved for rejected applications.

Features are recorded amount, debt-to-income ratio (DTI), FICO credit-score interval midpoint or recorded risk score, employment years, state, three-digit ZIP prefix, and normalized purpose; exclude accepted-only pricing/repayment fields, policy code, and calendar month. Within each training sample, median-impute numeric fields with missingness indicators, most-frequent-impute categorical fields, and one-hot encode categories with minimum frequency 25. Standardize numeric fields for logistic regression. Preprocessing, logistic regression, and stratified splitting use scikit-learn [26].

Eligibility, measurement, and extrapolation. The 2012 FICO minimum was 660 [19]. Source checks identify 180,903 below-660 applications (46.31%) and 23,097 missing standardized scores, with no issued loans in either group. All 53,367 issued loans lie among the remaining 186,644 records at or above 660. This structural restriction does not make FICO eligibility sufficient for joint covariate support or other underwriting requirements.

Recorded fields also differ [18]: rejected applications record FICO, whereas issued loans use an origination-score interval midpoint, without common coarsening; issued `loan_amnt` can reflect credit-department reductions, whereas rejected `Amount Requested` is the requested total. Issued purpose categories and rejected free-text titles are regex-normalized; dates refer to funding month and application date, respectively. Thus these are not identically measured predecision covariates.

In training fold $-k$, fit the issued-outcome regression $\widehat{r}_{-k}$ using the boosting settings below, evaluate every held-out record, and specify

$$r_i = \mathrm{clip}_{[.01,.99]}\, \widehat{r}_{-k(i)}(X_i), \qquad q_i \in [q_-(r_i), q_+(r_i)]. \tag{D.1}$$

On unsupported recorded covariates, r_i is an extrapolated reference, not an identified selected-label probability. Objectives use (5) with $p = r_i$, clipped fitted propensity, and the selected-observation residual correction. These residual-adjusted model-based estimates do not satisfy the population-overlap premise of Theorem 2 on the full cohort.

Models, cross-fitting, and optimization. Candidates use standard classification objectives: ℓ_2 logistic regression ($C = 1$, L-BFGS, 1000-iteration limit), default-weighted logistic regression (default class weight 2), and XGBoost [6] (450 trees, depth 5, learning rate .04, minimum child weight 10, row/column subsampling .85, ℓ_2 penalty 2, histogram trees, recorded CUDA configuration). Both nuisance classifiers use the boosting settings with separately fitted preprocessing and distinct seeds: p_1 uses revealed outcomes and π predicts S from all training covariates. Shuffled five-fold cross-fitting is stratified by S; clip $\widehat{p}_1$ to $[.01, .99]$ and $\widehat{\pi}$ to $[.02, .98]$. Evaluate Brier loss at $\alpha \in \{.90, .95\}$ and $\Gamma \in \{1, 1.25, 1.5, 2, 3, 5, 8, 12\}$.

The corrected objective can be nonconvex because hinge coefficients can be signed. Enumerate 0, 1, and both label-specific losses, $f(X_i)^2$ and $(1 - f(X_i))^2$, at every observed covariate X_i, including records whose actual outcomes are hidden. Deduplicate and sort increasingly; among exactly equal computed minima, choose the smallest threshold, without numerical tie tolerance. Clip endpoints to $[0, 1]$; if they cross, replace both by their midpoint. These clipping and crossing conventions are computational operations and do not supply statistical coverage.

Fold enlargement and diagnostics. At each endpoint's selected threshold, compute five fold-specific objectives and divide their sample standard deviation by $\sqrt{5}$ to obtain an empirical standard error. Let $t_{.975,4}$ be the .975 quantile of a Student-t distribution with four degrees of freedom. Subtract this multiplier times the lower-endpoint standard error from the raw lower endpoint and add the corresponding upper-endpoint quantity to the raw upper endpoint, before clipping or crossing repair. Then clip both limits to $[0, 1]$ and order them by minimum and maximum. This empirical diagnostic uses dependent fold-specific quantities because their training samples overlap; calibrated coverage requires the conditions and radius of Theorem 2.

Historical propensity AUC is .9944 and Brier score .0201; raw propensities are below .02 for 75.26% and above .98 for 1.80% of applicants, and inverse clipped propensities are at most 50, indicating substantial extrapolation. Subgroups reuse full-cohort cross-fitted predictions and require at least $\max\{250, \lfloor .002N \rfloor\}$ applications and 50 issued loans. At $\Gamma = 2, \alpha = .95$, boosting width is .080 for DTI above 30% (2.7% issued) and .032 for FICO 700–739 (30.3% issued).

α	Predictor	Accepted-only	$\Gamma = 1$	$\Gamma = 2$	$\Gamma = 3$
.95	Gradient boosting	.795	.733	[.703,.759]	[.682,.773]
.95	Logistic	.801	.805	[.770,.831]	[.745,.843]
.95	Default-weighted	.661	.813	[.782,.833]	[.757,.841]
.95	Constant $1/2$	.250	.250	[.250,.250]	[.250,.250]
.90	Gradient boosting	.740	.679	[.625,.715]	[.575,.732]
.90	Logistic	.741	.721	[.669,.760]	[.631,.778]
.90	Default-weighted	.579	.720	[.679,.747]	[.646,.759]
.90	Constant $1/2$	.250	.250	[.250,.250]	[.250,.250]

Table 2: Historical applicants: accepted-only CVaR and model-based applicant-level sensitivity intervals. Values other than the exact constant reference are estimates. At $\Gamma = 1$ the reference model fixes the hidden probabilities. Default-weighted denotes logistic regression with default class weight 2.

For $f \equiv 1/2$, both label-specific Brier losses equal $1/4$. Mean loss and every CVaR endpoint therefore equal $1/4$, and $d_{f,\eta} = \mathrm{TIV} = 0$ for every threshold. This row provides an exact predictor reference; the acquisition policies compare audits for a fixed or sequentially refitted boosting predictor.

Off-grid breakdown. For an accepted-only preferred f and comparator g, set $G(\Gamma) = \widehat{\rho}^{L}_{\Gamma}(g) - \widehat{\rho}^{U}_{\Gamma}(f)$. At the first grid value Γ_+ with $G(\Gamma_+) \le 0$, if its predecessor Γ_- has positive gap, interpolate:

$$\log \widehat{\Gamma}_{\mathrm{br}} = \log \Gamma_- + \frac{G(\Gamma_-)}{G(\Gamma_-) - G(\Gamma_+)} (\log \Gamma_+ - \log \Gamma_-).$$

These values interpolate the recorded grid. Boosting versus logistic gives approximate breakdown 2.24 at $\alpha = .95$, with first failed grid value 3, and 1.40 at $\alpha = .90$, with first failed value 1.5. Default-weighted versus logistic gives approximately 1.01 at $\alpha = .90$, with first failed value 1.25. The other three accepted-only orderings fail already at $\Gamma = 1$. A comparison with no crossing through 12 is recorded as right-censored above the grid.

D.4 Controlled masking: empirical coverage and model comparisons

For the 53,367 mature issued loans with known outcomes, independently draw 20 masks from

$$\begin{aligned} \operatorname{logit} P(S = 1 \mid X, Y) &= c + g(X) - Y \log 3, \\ g(X) &= \operatorname{clip}_{[-3,3]}\big\{.65z(\text{score}) - .35z(\text{DTI}) \\ &\qquad + .15z(\log(1 + \text{amount}))\big\}. \end{aligned}$$

Each z standardizes the median-imputed pseudo-population feature. Solve for expected revealed share .45 ($c = -.0423744$). Bayes' rule gives $\operatorname{odds}(p_0)/\operatorname{odds}(p_1) = 3$, hence $\Gamma_0 = 3$; boundary outcome probabilities coincide across hidden/revealed groups as in Appendix A.1. This validates artificial masking of mature issued loans, not outcomes for historically rejected applicants.

All three predictors share shuffled five-fold splits stratified by imposed S; fitting, clipping, threshold search, and fold enlargement follow Appendix D.3. Reference CVaR uses every known outcome with that predictor's cross-fitted predictions. Table 3 counts intervals containing this full-outcome reference risk. Table 4 checks strict separation of plug-in intervals against reference risk orderings. Three unordered predictor pairs per mask and 20 masks give 60 comparisons at each tail level, or 120 across both levels.

Figure 7 retains the complete assumed-Γ grid; plug-in coverage is zero at every tested $\Gamma \le 2$. Main-figure pooling averages 20 mask-level means over three predictors, giving 60 equally weighted predictor–mask rows from 20 independent masks. Fold enlargement is an empirical calibration diagnostic; the class-wide calibrated statement is Theorem 2.

α	Predictor	Plug-in	Enlarged	Mean width	Enlarged width
.90	Gradient boosting	11/20	19/20	.401	.434
.90	Logistic	13/20	20/20	.401	.434
.90	Default-weighted	14/20	20/20	.332	.365
.95	Gradient boosting	14/20	19/20	.070	.089
.95	Logistic	12/20	20/20	.066	.084
.95	Default-weighted	14/20	20/20	.110	.139

Table 3: Controlled masking at $\Gamma = \Gamma_0 = 3$. The middle columns count intervals covering the full pseudo-population reference CVaR over 20 masks; the last two columns are mean widths. Fold enlargement follows the empirical diagnostic in Appendix D.3.

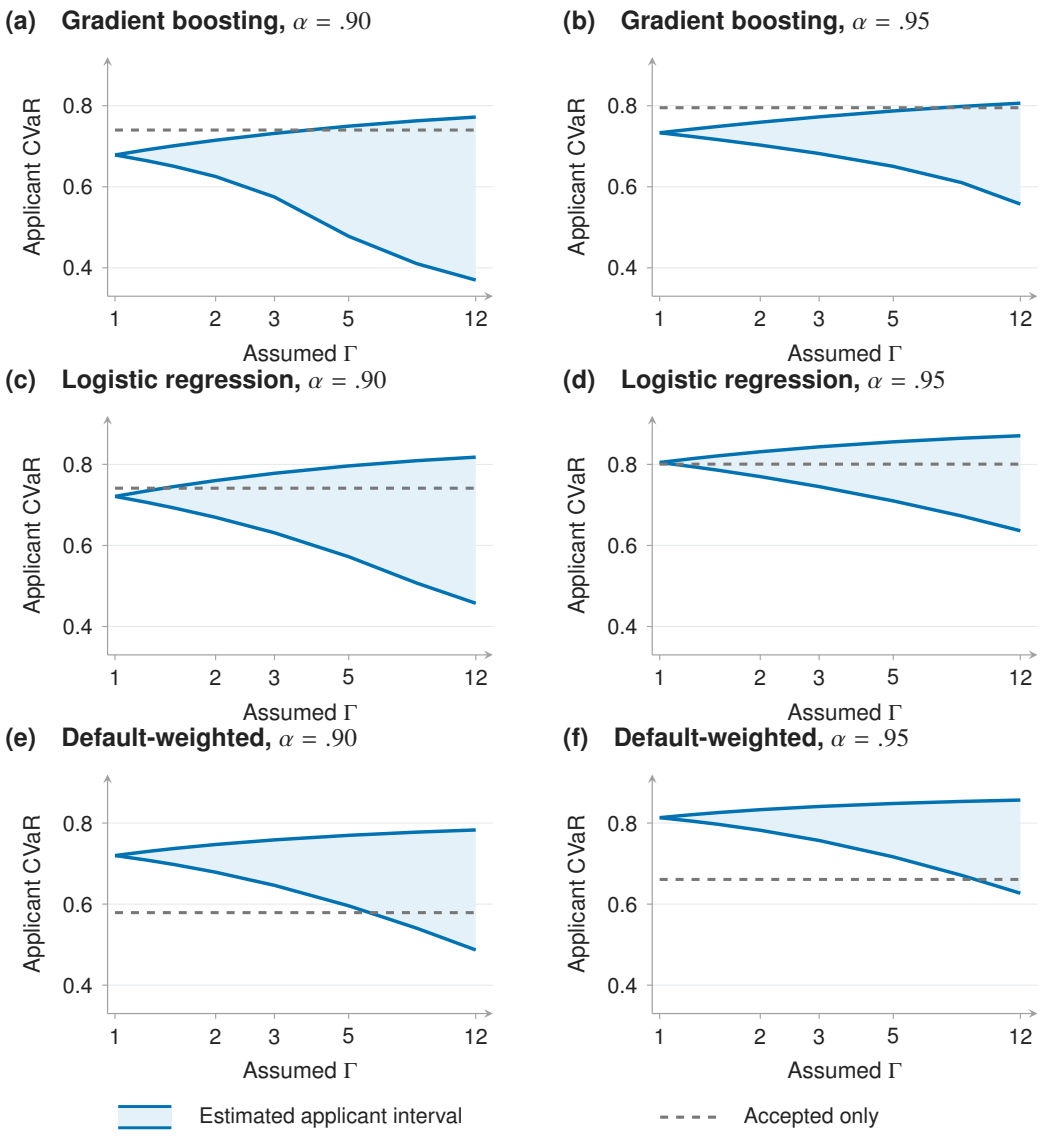


Figure 6: **Complete applicant-level sensitivity analysis.** All three predictors at $\alpha = .90$ and .95 over eight recorded sensitivity levels. Blue boundaries and shading show estimated sensitivity endpoints under the applicant regression extension of Appendix D.3. Gray dashed lines show accepted-only CVaR. Horizontal axes are logarithmic and vertical limits are shared. Default-weighted denotes default-weighted logistic regression.

Γ	α	Comparisons	Resolved	Incorrect	Unresolved
3	.90	60	0	0	60
3	.95	60	37	0	23
1	Both levels	120	120	16	0

Table 4: Model comparisons under controlled masking. Strict separation of plug-in intervals declares an ordering, whose correctness is checked against known-outcome reference CVaRs. Counts give resolved pairs, incorrect declared orderings, and unresolved pairs. Each tail level includes three unordered predictor pairs in each of 20 masks.

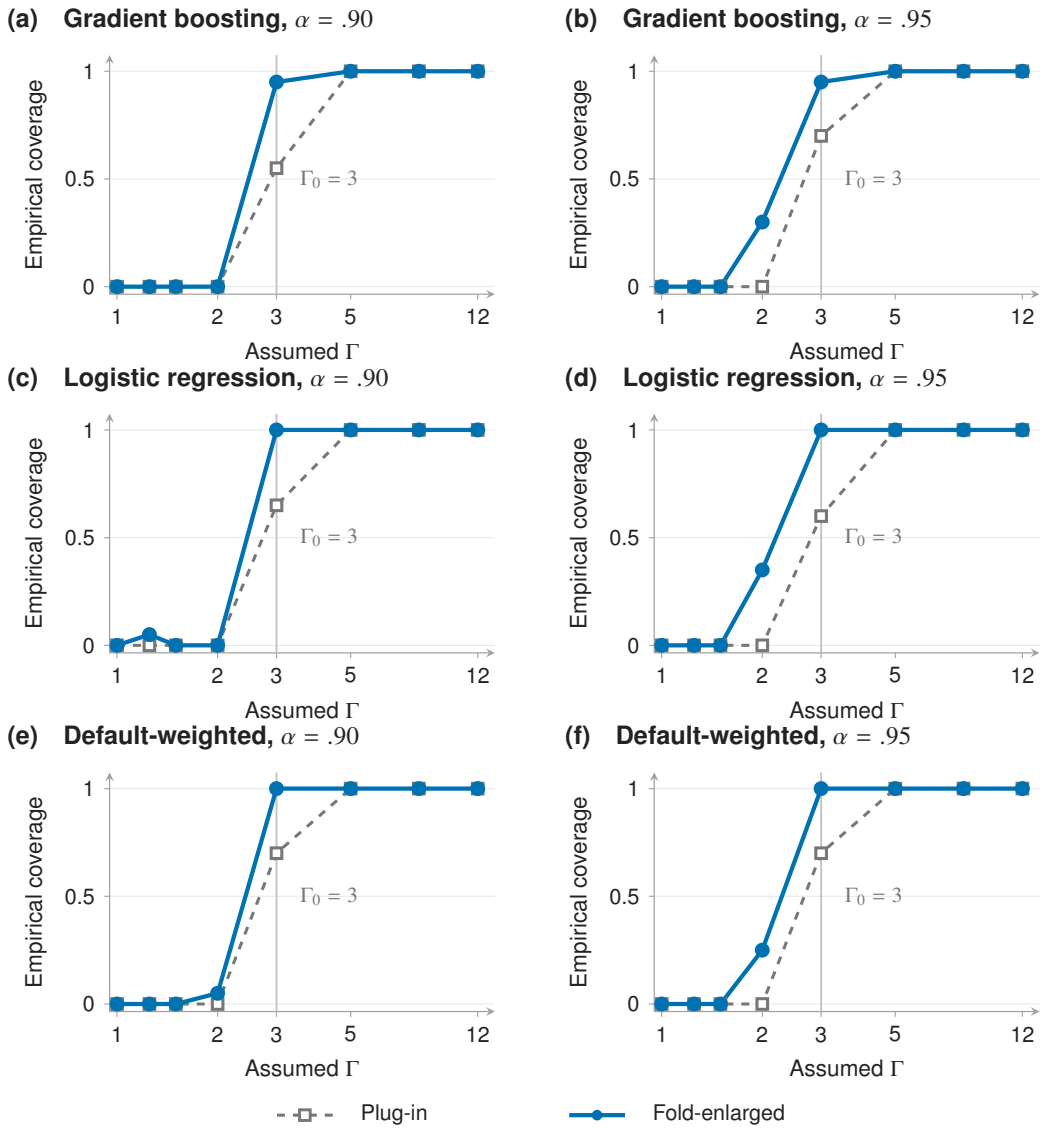


Figure 7: **Empirical coverage by predictor and tail level.** Each proportion uses 20 controlled masks. Blue circles show fold-enlarged intervals and hollow gray squares plug-in intervals; the vertical reference marks the masking mechanism's sensitivity $\Gamma_0 = 3$. Predictors share the same 20 masks, so the predictor-specific proportions are dependent. Default-weighted denotes default-weighted logistic regression.

D.5 Synthetic fixed-threshold cost–precision frontier

Generator. Use 120 worlds of 120 cells, $\alpha = .85$, $\Gamma = 12$, absolute loss, $f = p_1$, and $\eta = .5$. Masses are Dirichlet$(.75, \ldots, .75)$. Independently draw $z_j, e_j, e_j^{\text{out}}, e_j^{\text{in}} \sim N(0,1)$ across cells/noise types. Hidden-tail cells satisfy $z_j + .35e_j$ at or above its empirical .65 quantile (NumPy default linear interpolation). Outside/inside these cells, respectively, set

$$p_{1j} = \text{sigmoid}(-1.35 + .75z_j + .45e_j^{\text{out}}), \qquad p_{1j} = \text{sigmoid}(-2.05 + .35z_j + .30e_j^{\text{in}}),$$

clipped to $[.04, .72]$, and draw $\pi_j \sim U[.62, .95]/U[.02, .16]$. Add to the log-odds of p_{1j} a shift $N(0, .18^2)/U[.8 \log 10, \log 10]$ to obtain p_{0j}, clipped to $[.02, .98]$. Costs are $c_j = U_j + B_j V_j$ with $U_j \sim U[.45, 1.75]$, $B_j \sim \text{Bernoulli}(.20)$, and $V_j \sim U[.75, 1.75]$. Primitive draws are mutually independent and independent of masses; hidden-cell membership induces the specified dependence of π_j and log-odds shifts on the normal scores.

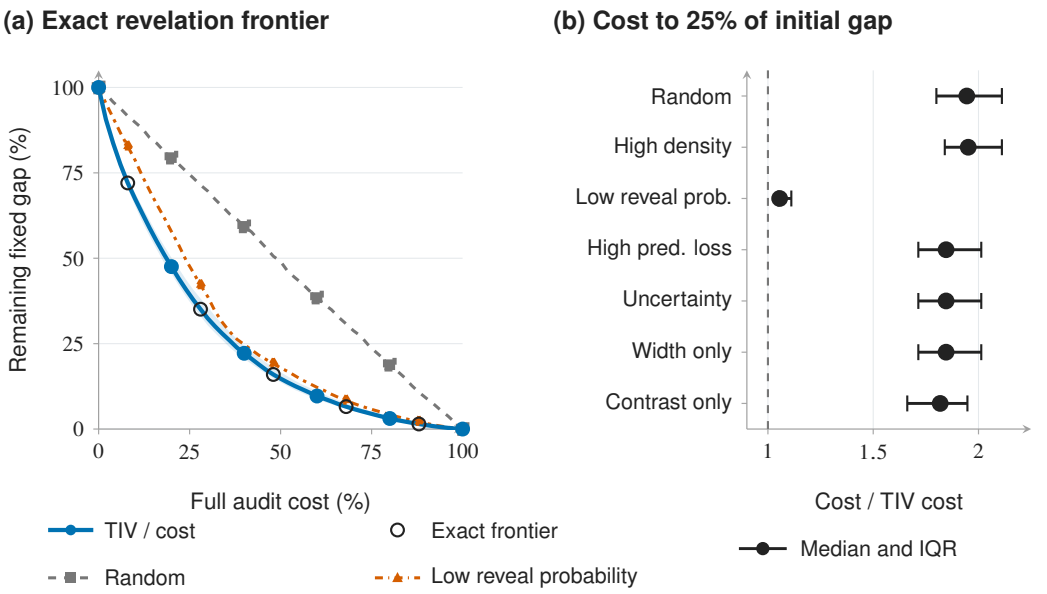


Figure 8: **The exact revelation frontier and the cost of alternative policies.** Across 120 finite-cell worlds, (a) median remaining fixed-threshold ambiguity, normalized within each world; blue shading is the TIV interquartile range and open markers identify the coincident exact frontier. (b) All seven baselines: median cost of reaching 25% of the initial gap relative to TIV in the same world, with interquartile bars. Costs use the recorded 2%-spaced grid.

The generated p_{0j} specifies the hidden-outcome distribution in cell j. The fixed-threshold frontier follows from $W_{f,\eta}(\lambda) = \sum_j \mu_j(1-\lambda_j)\tau_{f,\eta}(j)$ and cost $\sum_j \mu_j c_j \lambda_j$. The frontier calculation uses $p_1, \pi, f, \eta, \Gamma, \mu, c$; the generated p_0 does not enter these formulas.

Allocations and equivalent baselines. Here c_j is population cost density: allocations cost $\sum_j \mu_j c_j \lambda_j$, with full cost $\sum_j \mu_j c_j$. Appendix C.5's fractional-knapsack solution uses useful zero-cost cells first, then decreasing τ_j/c_j on positive-cost/value cells (this generator has positive costs). A fractional final cell implements 51 budgets from zero to full cost in .02 increments. TIV/cost and the exact frontier share this ordering and all saved coordinates.

Seven baselines rank by raw independent $U[0,1]$ random score, cell mass, negative revelation probability, selected-label expected absolute loss, $p_1(1-p_1)$, sensitivity width, or absolute tail-excess contrast $|d_{f,\eta}|$. Only TIV/frontier divide their scores by cost. Each world's random scores are independent of generator draws and reused across budgets. Decreasing NumPy default sorting (`argsort(score)[::-1]`) imposes no stable secondary tie rule, including clipping-induced ties. For $v_j = p_{1j}(1-p_{1j})$, predicted loss is $2v_j$, predictive uncertainty is v_j, and

$$s_\Gamma(p_{1j}) = \frac{(\Gamma^2-1)v_j}{\Gamma + (\Gamma-1)^2 v_j}. \tag{D.2}$$

The last expression increases strictly in v_j for $\Gamma > 1$, so these three baselines share a mathematical weak ordering and their saved curves coincide.

Normalize remaining ambiguity by its initial value within each world before taking medians/IQR. Cost-to-target is the first recorded budget reaching 25% of initial ambiguity, divided by TIV's corresponding cost in that world; there is no interpolation between budgets. Normalization requires positive initial ambiguity, full cost, and TIV target cost. This measures divisible population revelation at fixed f, η, Γ (Theorem 3).

D.6 Fixed and sequential audits of known repayments

Both protocols use boosting, $\alpha = .95$, $\Gamma = 3$, the 20 controlled masks, and unit cost per audited label. Let $N^{\text{fix}}_{\text{hidden,initial}}$ count initially hidden cases in the fixed protocol's original pool, and let $N^{\text{seq}}_{\text{hidden,initial}}$ count initially hidden cases in the sequential protocol's 80% training pool. Each protocol acquires eight batches of $\lceil .02 N^{\text{protocol}}_{\text{hidden,initial}} \rceil$ labels, capped by the remaining eligible pool. Its budget fraction is the cumulative acquired count divided by that same initial hidden count. Among candidates already known to be hidden, operational TIV is $\widehat{\tau}/(1-\widehat{\pi}) = \widehat{s}_\Gamma |d_{f,\eta}|/\beta$, corresponding to $c = 1 - \pi$ in Appendix C.5.

Protocol	Fixed objective	Sequential retraining
Data and predictor	Original hidden pool; initial cross-fitted predictor and historical nuisance predictions held fixed; upper threshold minimizes the empirical upper-endpoint objective in (6), built from the score (5), before auditing.	80/20 split jointly stratified by (S, Y); acquire only in training pool, refit predictor on all known training labels each round
Evaluation	Empirical mean of remaining-hidden indicators times original per-label TIV; normalize by round zero within each mask	Training-pool retained plug-in width; held-out mean Brier and CVaR on the same reserved 20%
Ordering/ties	One stable decreasing-score ordering per policy; ties retain masked-data order	Partial partition (`argpartition`); no stable secondary batch-cutoff rule; random scores refreshed each round

The fixed protocol takes its upper threshold from the empirical upper-endpoint objective in (6), built from the score (5), using the initial masking predictions of Appendix D.4 and the breakpoint/tie rule of Appendix D.3. It freezes those predictor and nuisance predictions, as well as this threshold, for every acquisition and fixed-gap calculation.

Retained objective. Historical p_1 is fitted on initially revealed training outcomes, clipped as in Appendix D.3, and frozen. For training size n, let $A_{it} \le 1 - S_i$ indicate audit by round t, f_t be the refitted predictor, and $\widehat{p}_i$ the frozen fit. The retained plug-in objective is

$$\widehat{Q}^\sigma_t(\eta) = \eta + \frac{1}{n\beta} \sum_{i=1}^{n} \Big[S_i a^{Y_i}_{f_t,\eta}(X_i) + A_{it} a^{Y_i}_{f_t,\eta}(X_i) + (1 - S_i - A_{it}) b^\sigma_{f_t,\eta}(X_i; \widehat{p}_i) \Big], \qquad \sigma \in \{L, U\}. \tag{D.3}$$

Known outcomes enter directly; still-hidden outcomes retain historical sensitivity envelopes. Each endpoint uses breakpoint minimization, the threshold tie rule, clipping, and midpoint collapse of Appendix D.3. At full revelation $A_{it} = 1 - S_i$, both objectives equal $\eta + (n\beta)^{-1} \sum_i a^{Y_i}_{f_t,\eta}(X_i)$ and both endpoints equal full-training-data empirical CVaR; a regression test verifies this identity. These training-prediction widths and held-out predictive risks are descriptive trajectories, without fold enlargement or adaptive confidence guarantees.

Acquisition rules. Among still-hidden candidates, let $\widehat{p}_{\text{known}}$ denote the fitted outcome probability from currently known labels, and let $\widehat{s}_\Gamma = q_+(\widehat{p}_1) - q_-(\widehat{p}_1)$ be the sensitivity width from the frozen historical outcome fit. In the sequential protocol, let $\widehat{\eta}^U$ be the current upper-endpoint minimizing threshold of (D.3). The six acquisition scores are TIV per label cost $\widehat{s}_\Gamma |d_{f,\widehat{\eta}^U}|/(\beta\kappa)$, with $\kappa = 1$; independent uniform random scores; predictive uncertainty $1 - 2|f - .5|$; negative fitted probability of currently having a known label; predicted Brier loss $(1 - \widehat{p}_{\text{known}}) f^2 + \widehat{p}_{\text{known}} (1 - f)^2$; and historical sensitivity width $\widehat{s}_\Gamma$. The fixed protocol uses its frozen initial predictor, threshold, and historical fits. Sequential TIV uses the current predictor and upper threshold with the frozen historical sensitivity bounds. The propensity baseline refits the indicator of a currently known label on all training covariates, with probabilities clipped to $[.02, .98]$. Predicted loss refits the outcome model on currently known outcomes and uses its raw probabilities. These models share the boosting settings above, with their respective targets, available labels, and seeds. Refreshed acquisition fits guide label choice; retained evaluation continues to use the historical outcome fit in (D.3).

Acquisition rule	Fixed-threshold gap reduction	Final median sequential interval width
TIV per label cost	61%	.035
Random	16%	.067
Lowest revelation propensity	9%	.084
Predicted loss	0%	.116
Predictive uncertainty	0%	.116
Sensitivity width	0%	.116

Table 5: Complete fixed and sequential audit comparison at the nominal 16% budget. Fixed reductions are medians of within-mask percentage reductions; estimated sequential widths are medians across the 20 runs. Integer rounding gives actual median budget fractions of 16.013% of the fixed initial hidden pool and 16.018% of the sequential initial hidden training pool.

Acquisition rule	Estimated width	Mean Brier	CVaR
TIV	0.0348	0.1362	0.8541
Random	0.0671	0.1347	0.8682
Paired TIV−Random	−0.0332	+0.0014	−0.0138
Exact predictor reference			
Constant $f \equiv 1/2$	0.0000	0.2500	0.2500

The constant forecast supplies an exact predictor reference.

Table 6: Precision and predictive loss after eight equal-budget audit rounds. Policy rows are medians across 20 masks; the paired row is the median of the within-mask TIV-minus-random differences. Estimated width uses the training pool; mean Brier loss and CVaR use the same reserved 20% split.

In all 20 paired terminal runs, TIV has lower estimated width and held-out CVaR than random, while random has lower mean Brier loss. The median paired differences are $-.0332$ in width, $-.0138$ in CVaR, and $+.0014$ in mean Brier loss (Table 6). The exact constant reference has lower tail loss and higher mean loss than both policy medians. Figure 9 shows how the estimated width and held-out tail risk evolve as each policy changes the training labels and predictor.

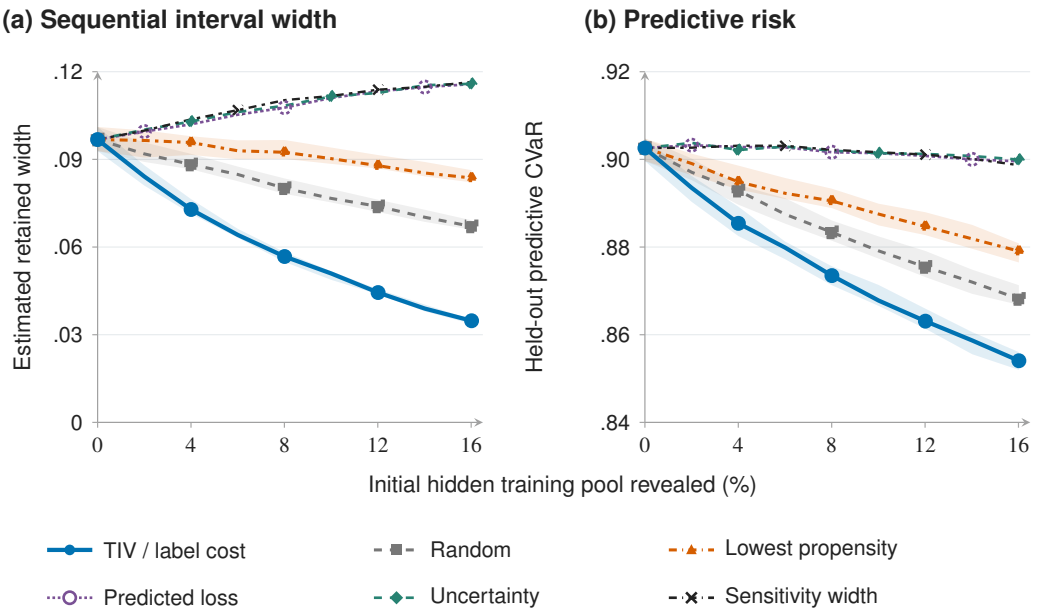


Figure 9: **Sequential precision and predictive outcomes across all six acquisition rules.** Lines are medians over 20 masks at $\alpha = .95$, $\Gamma = 3$; light bands are interquartile ranges for TIV, random, and lowest-propensity acquisition. (a) Estimated retained-sensitivity width for the current predictor. (b) Its CVaR on the reserved 20% split. Predicted-loss, predictive-uncertainty, and sensitivity-width curves are plotted separately and nearly overlap. Lines join recorded rounds. Table 6 gives the exact constant-forecast reference. The horizontal coordinate is the median fraction of initially hidden training-pool labels acquired.

D.7 Recorded implementation inventory

The accompanying `reproducibility/README.md` and `reproducibility/implementation_inventory.tex` record implementation entry points, configurations, seeds, draw order, saved-output mappings, and runtime/hash logs. The sequential results in Figure 4b, Figure 9, and Tables 5–6 use the retained plug-in objective (D.3); the inventory identifies its runner and configuration and records residual-corrected trajectories separately. It also documents the cohort verification: all 53,367 reconstructed issued records match all 12 shared columns in their original order, and all 20 archived mask diagnostics match.

E Sensitivity Calibration and Extensions

E.1 Choosing and calibrating the sensitivity range

For interior probabilities, $\gamma = \log \Gamma$ bounds the hidden–selected log-odds discrepancy:

$$\begin{aligned} |\operatorname{logit} p_0(x) - \operatorname{logit} p_1(x)| &\le \gamma, \\ p_0(x) &\in [\operatorname{expit}\{\operatorname{logit} p_1(x) - \gamma\}, \operatorname{expit}\{\operatorname{logit} p_1(x) + \gamma\}]. \end{aligned} \tag{E.1}$$

Overlap identifies the reference p_1; substantive knowledge or additional outcomes inform the allowed discrepancy, as in sensitivity analysis for nonignorable observation [30–32, 35].

Probability-scale elicitation. For a reference probability $p \in (0, 1)$ and elicited hidden range $[l, u] \subset (0, 1)$, the smallest symmetric odds bound containing that range is

$$\Gamma_{\text{elic}}(p, [l, u]) = \max\left\{1, \frac{\operatorname{odds}(u)}{\operatorname{odds}(p)}, \frac{\operatorname{odds}(p)}{\operatorname{odds}(l)}\right\}, \qquad \operatorname{odds}(v) = \frac{v}{1 - v}. \tag{E.2}$$

For $p = .10$ and $[l, u] = [.05, .20]$, this gives $9/4$. A common Γ must cover the whole target covariate support; uncertainty in p can be included by maximizing over its confidence set.

Pilot calibration and covariate resolution. Declare a finite-cell model $X = g \in \mathcal{G}$, and write $p_{sg} = \Pr(Y = 1 \mid X = g, S = s)$ in cells with originally hidden mass. A pilot identifies p_{0g} only when its design identifies that original hidden distribution: conditional random sampling suffices, while unequal probabilities require adjustment. An arbitrarily selected audited subset instead identifies its own law p_A (Appendix C.1). Suppose confidence sets satisfy $\Pr\{p_{sg} \in C_{sg} \text{ for all relevant } (s, g)\} \geq 1 - \delta_{\text{cal}}$. For sets contained in $(0, 1)$, define

$$\Gamma_{\text{cal}} = \exp\left\{ \max_{g \in \mathcal{G}} \sup_{\substack{u \in C_{0g} \\ v \in C_{1g}}} |\operatorname{logit} u - \operatorname{logit} v| \right\}. \tag{E.3}$$

On the simultaneous coverage event this bounds every cell's true discrepancy. The supremum can be infinite as admissible probabilities approach incompatible boundaries. Clipping confidence sets to an interior interval preserves coverage only under a justified probability restriction or a confidence construction already supplying it. The finite-Γ restrictions at $p_1 = 0, 1$ are specified in Appendix A.1.

Calibration in (E.3) controls the model with $X = g$. Cell averages do not control discrepancies at finer covariate resolution: either analyze with these cells or justify explicit within-cell homogeneity of the relevant probabilities. A certificate using finer covariates needs control at that finer resolution.

Joint error control. With an independent pilot, condition on it and apply the certificate at the chosen finite Γ_{cal} with valid conditional failure probability δ_{cert}, including nuisance estimation. A union bound gives total failure probability at most $\delta_{\text{cal}}+\delta_{\text{cert}}$ under the two procedures' assumptions. Reusing labels for calibration and revelation requires joint control of the sensitivity choice, audited contribution, and endpoint estimates; their dependence must enter the analysis. For a prespecified finite sensitivity grid, allocate error across its points, each already controlling the stipulated predictor class. A continuum requires uniform control in sensitivity. Let $\Gamma_{\max}$ be the largest sensitivity level under consideration. Population nesting implies that risk-limit or strict-separation claims valid at $\Gamma_{\max}$ hold throughout $[1, \Gamma_{\max}]$. Compare the calibrated range with the breakdown definition in Appendix A.5; Appendix D.3 estimates breakdown numerically by interpolation on a log-sensitivity grid.

E.2 Extensions of the outcome model

Binary outcomes provide two label-specific losses whose ordering supplies one extremal hidden kernel for every threshold. A multiclass extension needs a sensitivity set for the hidden class-probability vector and a new extremality and threshold-optimization argument. Continuous outcomes require conditional-distribution sensitivity sets; unbounded losses additionally require tail conditions for finite statistical control. Population overlap and nuisance-error requirements remain those of Appendix B; the historical extension and masking calibration are specified in Appendices D.3 and D.4.

E.3 Connections to related approaches

Selective-label evaluation concerns outcomes observed through historical decisions [5, 9, 17]. Missing-data analysis and partial identification organize the resulting uncertainty through compatible laws [20–23]; here every point of the tail-risk interval has such a law. The CVaR representation accommodates discrete losses [1, 2, 28, 29]. As in distributionally robust learning [4, 11, 12, 25], the upper endpoint is a worst-case risk; the lower endpoint records the remaining identification information. Orthogonalization and cross-fitting control nuisance bias [7, 14, 27], and Rademacher complexity and contraction control uniform fluctuation [3, 24, 33].

Active learning links queries to a learning objective [8]; TIV links acquisition to the precision of a specified risk claim. Its exact allocation theorem fixes f, η, Γ. The sequential study (Appendix D.6) refreshes acquisition values after retraining while keeping the historical outcome regression fixed in the retained objective. Auxiliary fits to current known-label status or outcomes serve the corresponding acquisition baselines. The study evaluates interval trajectories and held-out predictive risk. Inference across adaptive rounds or at data-dependent stopping requires control of the acquisition history, newly revealed contributions, and stopping rule, beyond the fixed-sample uniform theorem.